

# Supplement B - Simulation Atlas

## Extended sensitivity analyses for The Link Function Problem in Psychological Interaction Testing

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## 1 Purpose and scope

This document reports the extended simulation atlas for *The Link Function Problem in Psychological Interaction Testing*. It follows the terminology and simulation numbering of the main manuscript. The manuscript keeps a small number of transparent worked scenarios. The atlas asks how far those results extend when selected design and data-generating parameters are varied around the manuscript anchors. It is designed to be read independently of the interactive application: the relevant design settings, coefficients, fitted models, and outcome summaries are reported here, while references to the manuscript identify the corresponding conceptual examples.

The manuscript-anchor scenarios use the same data-generating settings as the corresponding manuscript simulations, but they are independently rerun as part of the atlas. Their Monte Carlo rejection rates can therefore differ slightly from those reported in the manuscript and Supplement A, whereas the underlying generating parameters and deterministic scale contrasts are the same.

The atlas is a structured sensitivity analysis, not an exhaustive map of psychological settings. It uses prespecified one-dimensional slices and a small main-effect surface instead of a full Cartesian product. In every scenario, the interaction term is **exactly zero on the generating scale**. Therefore, a rejection under the matched generating model estimates the nominal generating-scale rejection rate, while a rejection under another scale is described as a **pseudo-interaction detection rate**. As in the manuscript, *pseudo-interaction* is a simulation term. It is available here because the generating scale is known.

The current compact summaries come from atlas run type full, atlas version 0.1.0. The atlas contains 93 unique generating scenarios: 24 for forced-choice accuracy, 46 for the logit-probit comparison, and 23 for sum scores. The forced-choice and sum-score scenarios use 3000 Monte Carlo replications per scenario; the more expensive within-family GLMM scenarios use 3000.

### 1.1 How to read the figures

All rejection-rate figures use the same interpretation. The horizontal dashed line at .05 marks nominal alpha. Rates close to .05 under the generating scale provide a calibration check. Elevated rates under an alternative fitted scale show how often the scale difference becomes a statistically significant product term at the stated sample size and design. Error bars are shown for the manuscript anchors. The sensitivity plots omit error bars to keep the larger

set of scenarios readable; Wilson 95% intervals are available in the numerical tables and in `data/atlas-summary.csv`.

The deterministic pseudo-interaction reported in the tables is computed from the expected values before sampling. It describes the interaction contrast implied by evaluating the generating pattern on the fitted scale. The Monte Carlo rejection rate adds sampling variability, standard errors, and sample size to that deterministic discrepancy.

## 2 Atlas design

Table 1: Structure of the simulation atlas. Each family contains the manuscript anchor scenarios plus prespecified sensitivity slices.

| Manuscript case                      | Generating scenarios | Monte Carlo replications | Sensitivity slices                                                       |
|--------------------------------------|----------------------|--------------------------|--------------------------------------------------------------------------|
| Simulation 1: Forced-choice accuracy | 24                   | 3000                     | Main effects; sample size; intercept; trials; chance level               |
| Simulation 2: Logit vs probit        | 46                   | 3000                     | Main effects; sample size; trials; intercept; ICC                        |
| Simulation 3: Sum scores             | 23                   | 3000                     | Main effects; sample size; scale location; item count; latent dispersion |

The one-dimensional slices change one named parameter while holding the remaining settings at the central manuscript anchor. The main-effect surfaces vary the two additive main-effect coefficients jointly. The actual coefficients are reported directly on the figure axes and in the table below. Duplicate anchor combinations are stored once and can belong to several slices.

Table 2: Generating-scale coefficients used in the main-effect surfaces. Coefficients are reported on the scale that generated each scenario.

| Manuscript case                      | Generating link/scale  | First main effect | First coefficient values | Second main effect | Second coefficient values |
|--------------------------------------|------------------------|-------------------|--------------------------|--------------------|---------------------------|
| Simulation 1: Forced-choice accuracy | Chance-corrected logit | Age coefficient   | 0.30, 0.60, 0.90         | Group coefficient  | -1.35, -0.90, -0.45       |

Table 2: Generating-scale coefficients used in the main-effect surfaces. Coefficients are reported on the scale that generated each scenario.

| Manuscript case               | Generating link/scale   | First main effect     | First coefficient values | Second main effect    | Second coefficient values |
|-------------------------------|-------------------------|-----------------------|--------------------------|-----------------------|---------------------------|
| Simulation 2: Logit vs probit | Logit                   | Group coefficient     | -2.475, -1.650, -0.825   | Condition coefficient | -2.475, -1.650, -0.825    |
| Simulation 2: Logit vs probit | Probit                  | Group coefficient     | -1.50, -1.00, -0.50      | Condition coefficient | -1.50, -1.00, -0.50       |
| Simulation 3: Sum scores      | Latent generating scale | Predictor coefficient | 0.425, 0.850, 1.275      | Group coefficient     | -1.35, -0.90, -0.45       |

For Simulation 2, logit and probit coefficients are on different generating-link scales. As in the manuscript simulation, the logit coefficients are scaled by 1.65 relative to the probit reference coefficients so that the two generating scenarios have similar cell probabilities. The figures therefore report the actual coefficient values for each generating link instead of a common relative index.

### 3 Simulation 1: Forced-choice accuracy and the non-zero chance floor

The manuscript generates accuracy from a chance-corrected binomial logit model with a .50 floor and no age-by-group product term on that scale. It compares the matched chance-corrected model with a standard binomial logit, a standard binomial probit, and a Gaussian identity model. The atlas keeps this comparison and extends the paper scenarios across sample size, overall performance, number of trials, chance level, and the joint values of the age and group coefficients. In the main-effect surface, the generating-scale age coefficients are 0.30, 0.60, 0.90 and the group coefficients are -1.35, -0.90, -0.45.

#### 3.1 Manuscript anchors

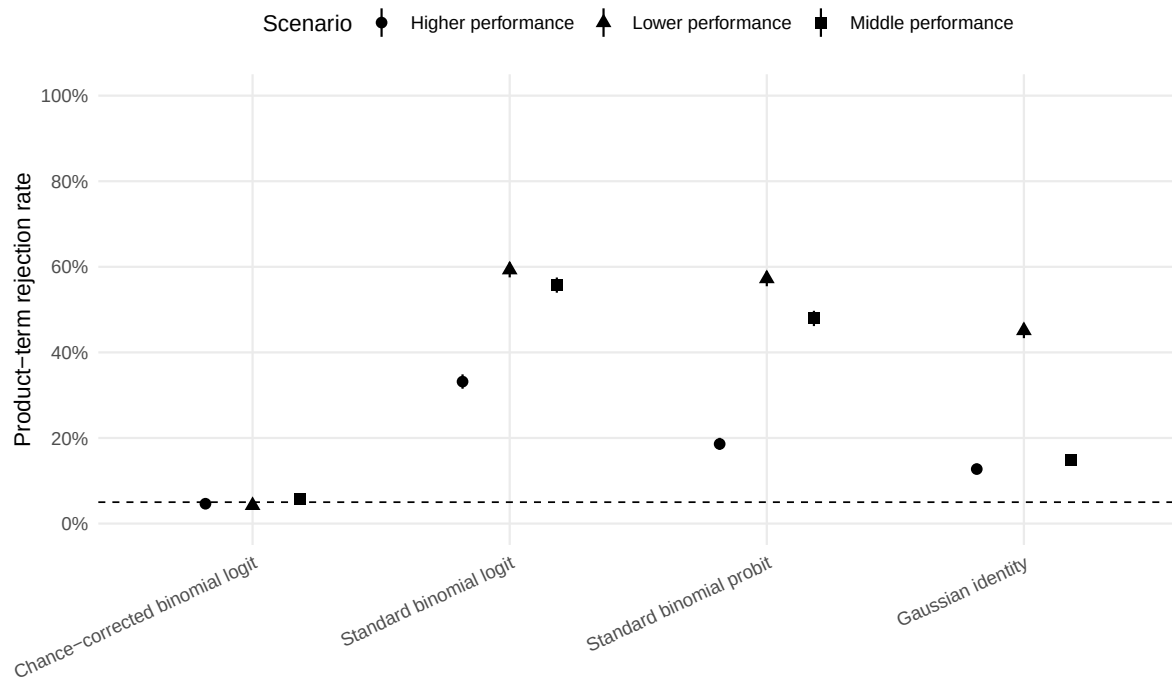


Figure 1: Forced-choice manuscript anchors reproduced within the atlas. The product term is zero on the chance-corrected logit generating scale. The dashed line marks  $\alpha = .05$ .

Table 3: Forced-choice manuscript anchors. The deterministic pseudo-interaction is evaluated on each fitted scale before sampling.

| Scenario           | Generating link/scale  | Fitted model/scale              | Rate type                               | Rejection rate | 95% CI         | Deterministic pseudo-interaction | Fit success |
|--------------------|------------------------|---------------------------------|-----------------------------------------|----------------|----------------|----------------------------------|-------------|
| Lower performance  | chance-corrected logit | Chance-corrected binomial logit | Nominal generating-scale rejection rate | 4.2%           | [3.6%, 5.0%]   | 0.000                            | 100.0%      |
| Lower performance  | chance-corrected logit | Gaussian identity               | Pseudo-interaction detection rate       | 45.1%          | [43.3%, 46.9%] | -0.077                           | 100.0%      |
| Lower performance  | chance-corrected logit | Standard binomial logit         | Pseudo-interaction detection rate       | 59.3%          | [57.5%, 61.0%] | -0.453                           | 100.0%      |
| Lower performance  | chance-corrected logit | Standard binomial probit        | Pseudo-interaction detection rate       | 57.2%          | [55.5%, 59.0%] | -0.262                           | 100.0%      |
| Middle performance | chance-corrected logit | Chance-corrected binomial logit | Nominal generating-scale rejection rate | 5.8%           | [5.0%, 6.7%]   | 0.000                            | 100.0%      |
| Middle performance | chance-corrected logit | Gaussian identity               | Pseudo-interaction detection rate       | 14.9%          | [13.7%, 16.2%] | -0.036                           | 100.0%      |
| Middle performance | chance-corrected logit | Standard binomial logit         | Pseudo-interaction detection rate       | 55.7%          | [53.9%, 57.5%] | -0.473                           | 100.0%      |
| Middle performance | chance-corrected logit | Standard binomial probit        | Pseudo-interaction detection rate       | 47.9%          | [46.1%, 49.7%] | -0.243                           | 100.0%      |
| Higher performance | chance-corrected logit | Chance-corrected binomial logit | Nominal generating-scale rejection rate | 4.6%           | [3.9%, 5.4%]   | 0.000                            | 100.0%      |
| Higher performance | chance-corrected logit | Gaussian identity               | Pseudo-interaction detection rate       | 12.7%          | [11.6%, 14.0%] | 0.028                            | 100.0%      |

Table 3: Forced-choice manuscript anchors. The deterministic pseudo-interaction is evaluated on each fitted scale before sampling.

| Scenario           | Generating link/scale  | Fitted model/scale       | Rate type                         | Rejection rate | 95% CI         | Deterministic pseudo-interaction | Fit success |
|--------------------|------------------------|--------------------------|-----------------------------------|----------------|----------------|----------------------------------|-------------|
| Higher performance | chance-corrected logit | Standard binomial logit  | Pseudo-interaction detection rate | 33.2%          | [31.5%, 34.9%] | -0.396                           | 100.0%      |
| Higher performance | chance-corrected logit | Standard binomial probit | Pseudo-interaction detection rate | 18.6%          | [17.2%, 20.0%] | -0.153                           | 100.0%      |

Across all forced-choice atlas scenarios, the matched chance-corrected model remains close to nominal alpha, with rejection rates from 3.6% to 5.8%. The standard binomial logit ranges from 8.0% to 99.0%, and the standard binomial probit from 6.3% to 97.2%. These ranges are sensitivity results, not prevalence estimates. They show that the inferential consequence of placing the lower asymptote at 0 instead of the task-implied chance level can be small in some regions and very large in others.

### 3.2 One-dimensional sensitivity slices

Table 4: Range of rejection rates across each forced-choice sensitivity slice.

| Slice                  | Generating link/scale  | Fitted model/scale              | Minimum rejection rate | Maximum rejection rate |
|------------------------|------------------------|---------------------------------|------------------------|------------------------|
| Chance level           | chance-corrected logit | Chance-corrected binomial logit | 4.6%                   | 5.8%                   |
| Intercept              | chance-corrected logit | Chance-corrected binomial logit | 4.2%                   | 5.8%                   |
| Sample size            | chance-corrected logit | Chance-corrected binomial logit | 4.3%                   | 5.8%                   |
| Trials per participant | chance-corrected logit | Chance-corrected binomial logit | 3.6%                   | 5.8%                   |
| Chance level           | chance-corrected logit | Standard binomial logit         | 47.5%                  | 55.7%                  |
| Intercept              | chance-corrected logit | Standard binomial logit         | 14.0%                  | 59.3%                  |

Table 4: Range of rejection rates across each forced-choice sensitivity slice.

| Slice                  | Generating link/scale  | Fitted model/scale       | Minimum rejection rate | Maximum rejection rate |
|------------------------|------------------------|--------------------------|------------------------|------------------------|
| Sample size            | chance-corrected logit | Standard binomial logit  | 15.2%                  | 99.0%                  |
| Trials per participant | chance-corrected logit | Standard binomial logit  | 18.0%                  | 90.5%                  |
| Chance level           | chance-corrected logit | Standard binomial probit | 44.7%                  | 48.2%                  |
| Intercept              | chance-corrected logit | Standard binomial probit | 6.3%                   | 57.2%                  |
| Sample size            | chance-corrected logit | Standard binomial probit | 13.8%                  | 97.2%                  |
| Trials per participant | chance-corrected logit | Standard binomial probit | 15.7%                  | 83.8%                  |
| Chance level           | chance-corrected logit | Gaussian identity        | 14.9%                  | 25.7%                  |
| Intercept              | chance-corrected logit | Gaussian identity        | 12.7%                  | 65.5%                  |
| Sample size            | chance-corrected logit | Gaussian identity        | 6.5%                   | 47.2%                  |
| Trials per participant | chance-corrected logit | Gaussian identity        | 7.1%                   | 30.5%                  |

The slices separate distinct reasons why a deterministic scale discrepancy may or may not become a statistical interaction. The intercept and the chance level change the curvature relevant to the fitted link. Sample size and number of trials change how precisely that discrepancy is estimated. The figure therefore distinguishes the existence of a pseudo-interaction on the fitted scale from the probability that a product-term test detects it.

### 3.3 Main-effect surface

The surface directly targets the mechanism emphasized in the manuscript. Two additive main effects can jointly move observations through regions where equal differences on the generating scale map to unequal differences on the fitted scale. Larger main effects can therefore increase the deterministic pseudo-interaction even though the generating age-by-group term remains zero.



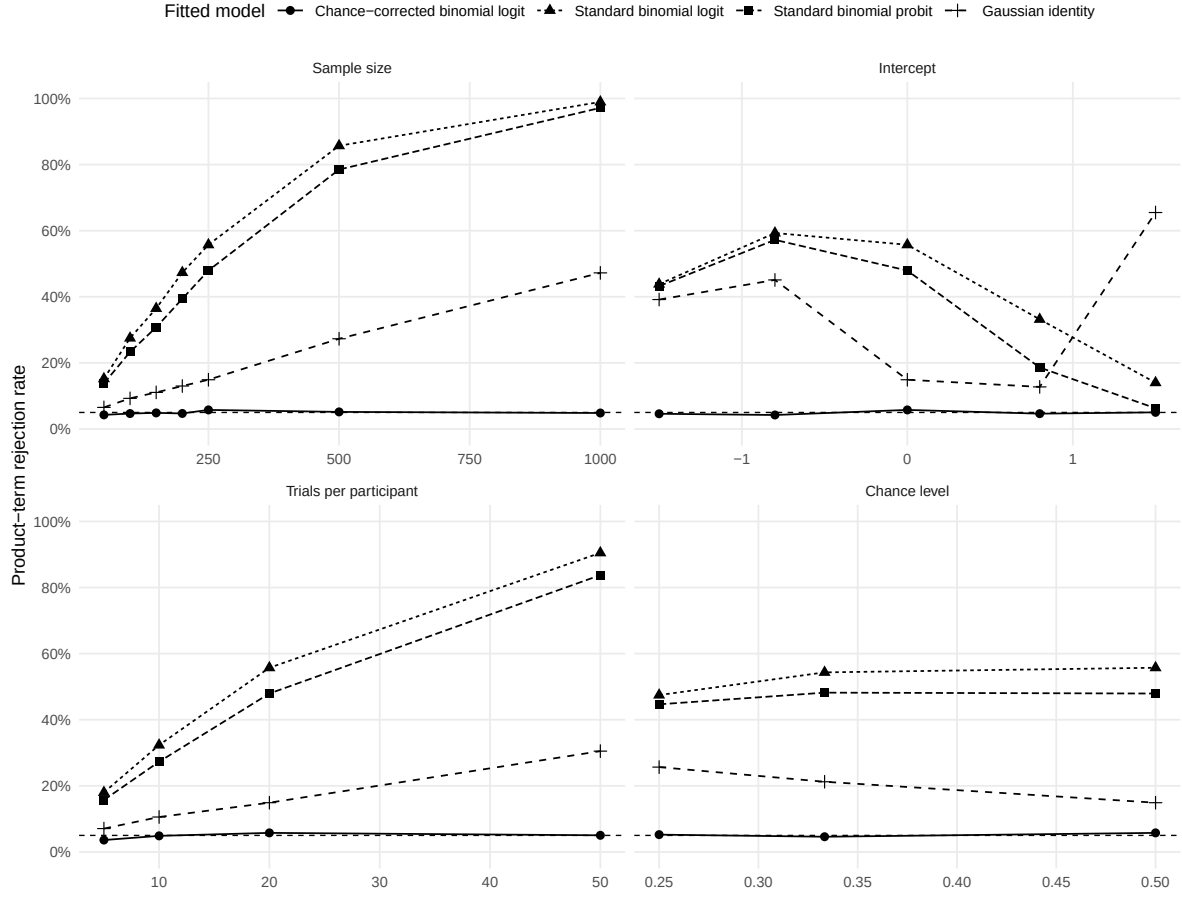


Figure 2: Forced-choice sensitivity slices. One parameter varies at a time around the central manuscript anchor. The dashed line marks  $\alpha = .05$ .

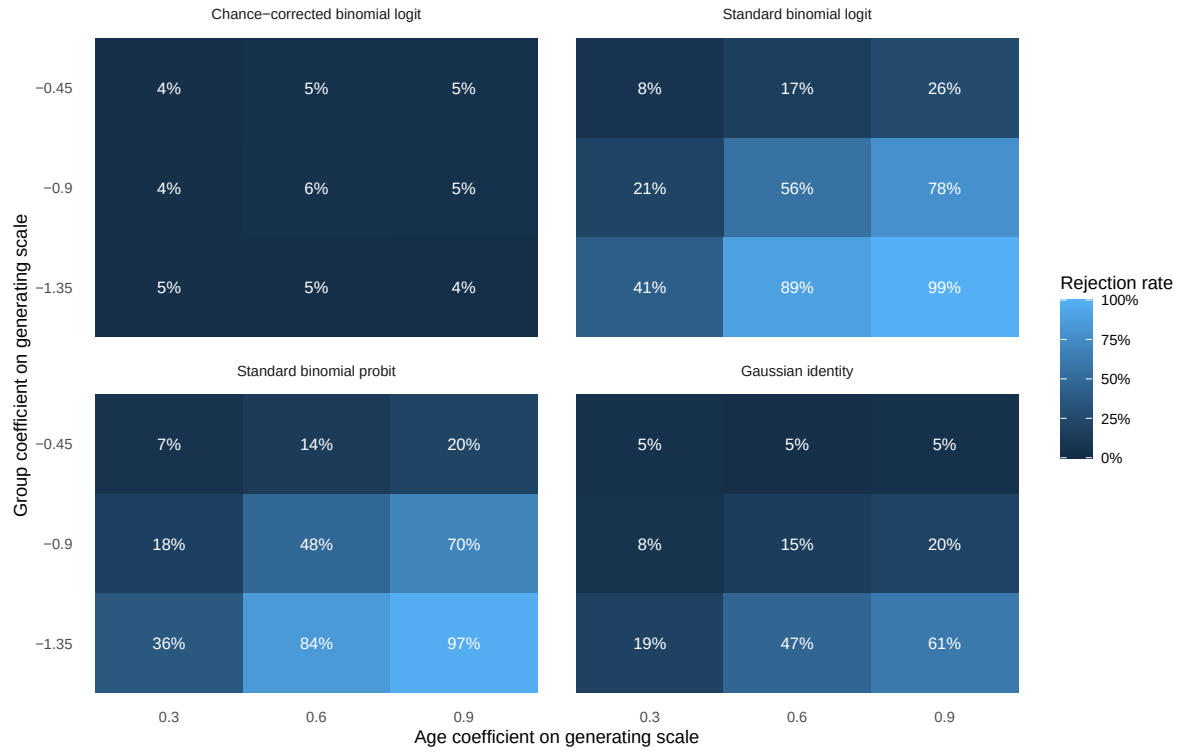


Figure 3: Forced-choice main-effect surface. Axes report the age and group coefficients on the chance-corrected logit generating scale. Cell labels are product-term rejection rates.

## 4 Simulation 2: Within-family link choice, logit versus probit

The manuscript crosses logit and probit as generating and fitted links in repeated-trial random-intercept GLMMs. Both links use the binomial family and the same 0-to-1 bounds. The difference is the scale on which additivity is imposed. The atlas extends this comparison across sample size, trials per participant-condition cell, the intercept, latent ICC, and the joint values of the group and condition coefficients. In the main-effect surface, the probit generating coefficients for both effects are -1.50, -1.00, -0.50, whereas the corresponding logit coefficients are -2.475, -1.650, -0.825.

### 4.1 Manuscript anchors

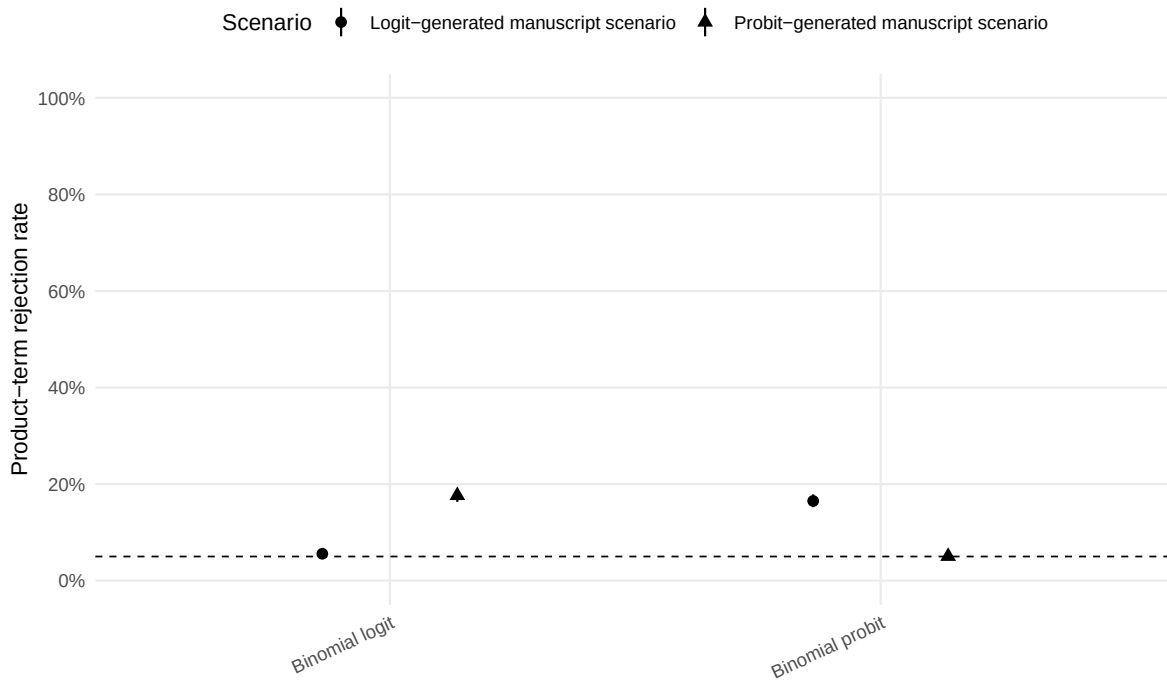


Figure 4: Logit-probit manuscript anchors reproduced within the atlas. The product term is zero on the generating link scale. The dashed line marks  $\alpha = .05$ .

Table 5: Within-family manuscript anchors for logit and probit generating links.

| Scenario                             | Generating link/scale | Fitted model/scale | Rate type                               | Rejection rate | 95% CI         | Deterministic pseudo-interaction | Fit success |
|--------------------------------------|-----------------------|--------------------|-----------------------------------------|----------------|----------------|----------------------------------|-------------|
| Logit-generated manuscript scenario  | logit                 | Binomial logit     | Nominal generating-scale rejection rate | 5.6%           | [4.8%, 6.4%]   | 0.000                            | 100.0%      |
| Logit-generated manuscript scenario  | logit                 | Binomial probit    | Pseudo-interaction detection rate       | 16.5%          | [15.2%, 17.9%] | -0.112                           | 100.0%      |
| Probit-generated manuscript scenario | probit                | Binomial logit     | Pseudo-interaction detection rate       | 17.6%          | [16.3%, 19.0%] | 0.216                            | 100.0%      |
| Probit-generated manuscript scenario | probit                | Binomial probit    | Nominal generating-scale rejection rate | 5.0%           | [4.3%, 5.9%]   | 0.000                            | 100.0%      |

Across the atlas, matched logit/probit fits remain near the nominal rejection rate. The wrong-link rows range from 4.4% to 82.3%. The range is wider than the manuscript anchor alone suggests because the atlas deliberately varies information and the region of the latent scale. The result remains conditional on these declared slices and should not be read as a general frequency for logit-probit disagreement.

## 4.2 One-dimensional sensitivity slices

Table 6: Range of rejection rates across each within-family sensitivity slice.

| Slice           | Generating link/scale | Fitted model/scale | Minimum rejection rate | Maximum rejection rate |
|-----------------|-----------------------|--------------------|------------------------|------------------------|
| Intercept       | logit                 | Binomial logit     | 5.1%                   | 5.6%                   |
| Latent ICC      | logit                 | Binomial logit     | 4.9%                   | 5.9%                   |
| Sample size     | logit                 | Binomial logit     | 4.4%                   | 5.7%                   |
| Trials per cell | logit                 | Binomial logit     | 4.4%                   | 5.6%                   |

Table 6: Range of rejection rates across each within-family sensitivity slice.

| Slice           | Generating link/scale | Fitted model/scale | Minimum rejection rate | Maximum rejection rate |
|-----------------|-----------------------|--------------------|------------------------|------------------------|
| Intercept       | probit                | Binomial logit     | 8.0%                   | 46.2%                  |
| Latent ICC      | probit                | Binomial logit     | 8.6%                   | 82.3%                  |
| Sample size     | probit                | Binomial logit     | 6.6%                   | 41.5%                  |
| Trials per cell | probit                | Binomial logit     | 7.9%                   | 32.2%                  |
| Intercept       | logit                 | Binomial probit    | 8.2%                   | 44.1%                  |
| Latent ICC      | logit                 | Binomial probit    | 9.2%                   | 78.7%                  |
| Sample size     | logit                 | Binomial probit    | 7.4%                   | 38.5%                  |
| Trials per cell | logit                 | Binomial probit    | 8.2%                   | 31.0%                  |
| Intercept       | probit                | Binomial probit    | 4.7%                   | 5.7%                   |
| Latent ICC      | probit                | Binomial probit    | 4.8%                   | 5.4%                   |
| Sample size     | probit                | Binomial probit    | 4.5%                   | 5.6%                   |
| Trials per cell | probit                | Binomial probit    | 4.6%                   | 5.0%                   |

Sample size and trial count mainly change how detectable the small systematic discrepancy between the links becomes. The intercept and the main-effect magnitudes change the discrepancy itself by moving cell probabilities into regions where logit and probit curvature differs more. ICC changes the information structure of the repeated observations while leaving the fixed generating interaction at zero.

### 4.3 Main-effect surface

The matched-link panels provide the calibration reference. The off-diagonal panels show when the small difference between two standard binomial links becomes consequential for an interaction claim. This is the narrow point of the within-family example: logit and probit can remain

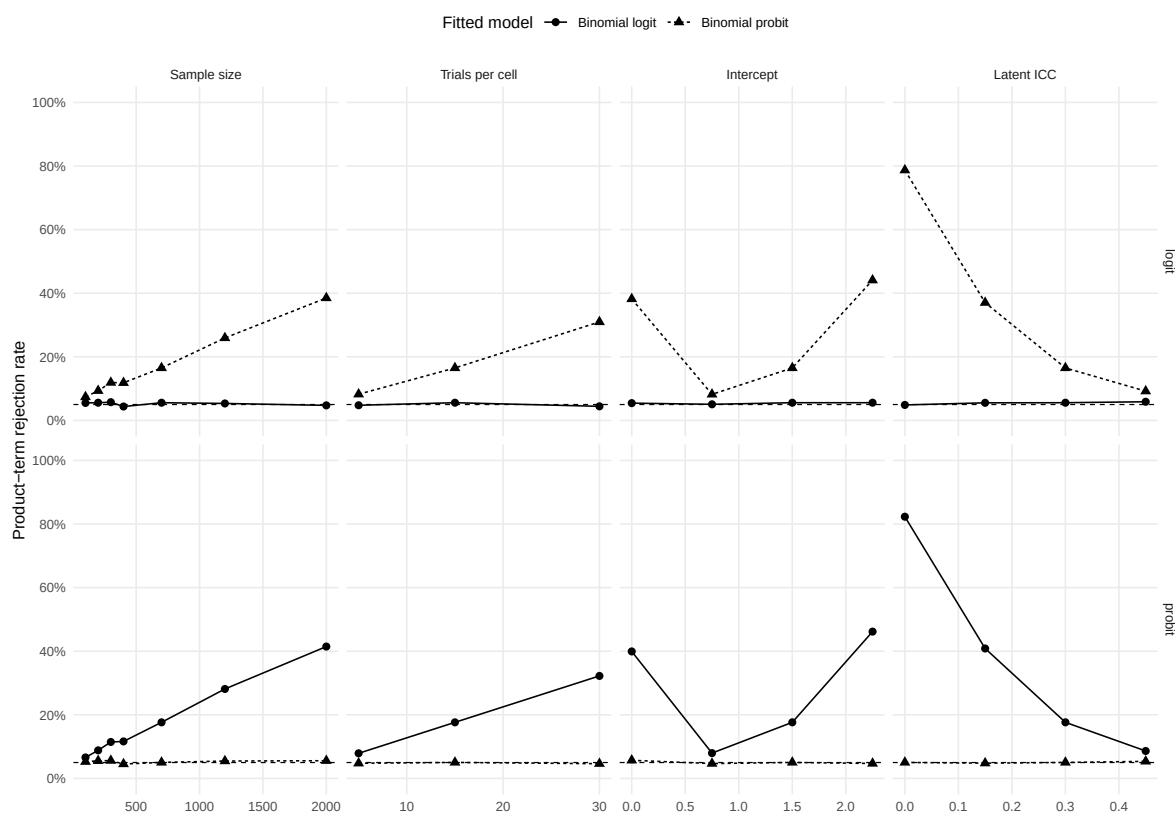


Figure 5: Logit-probit sensitivity slices. Rows identify the generating link. Within each panel the fitted logit and probit models are compared as one parameter varies around the manuscript anchor.

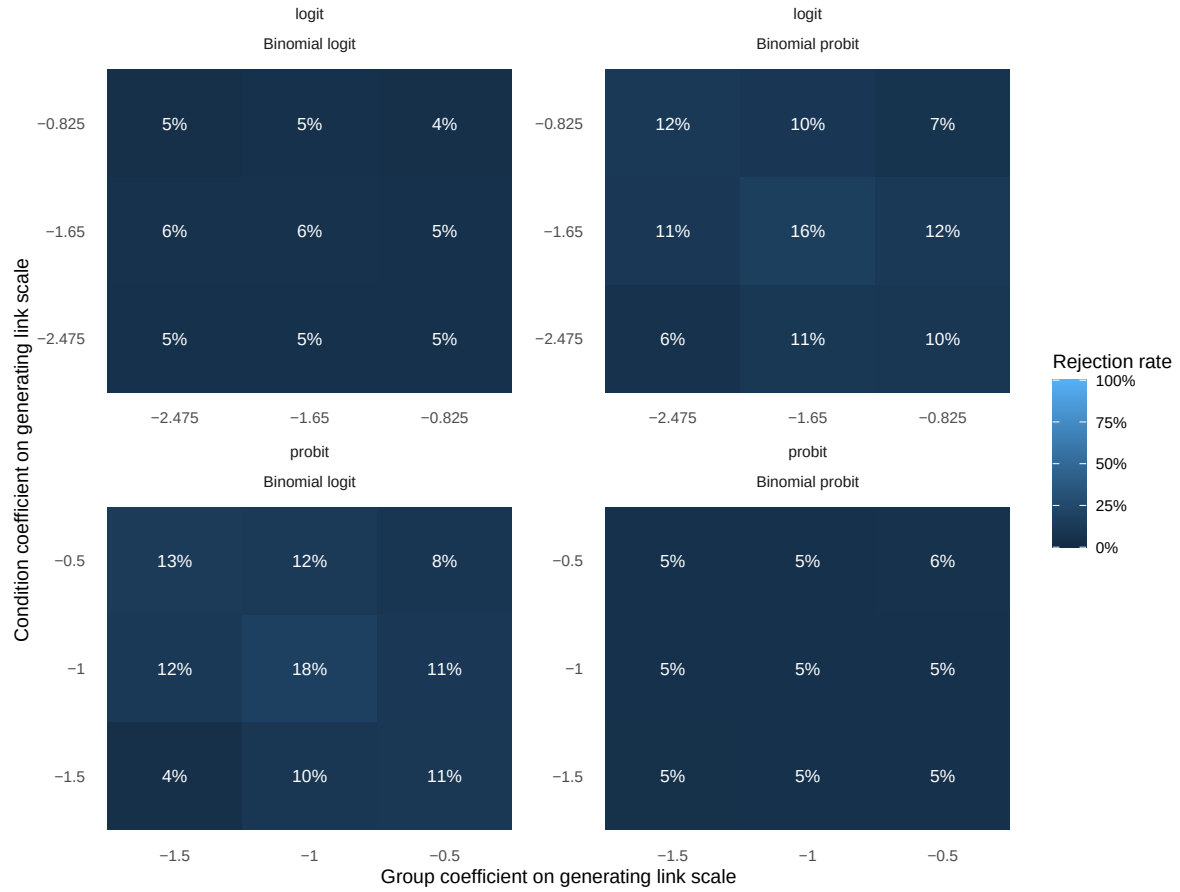


Figure 6: Within-family main-effect surfaces. Axes report the group and condition coefficients on each generating-link scale. Panels identify the generating and fitted links. Cell labels are product-term rejection rates.

very similar for prediction while defining sufficiently different additivity baselines to change a product-term conclusion under high information or strong curvature.

## 5 Simulation 3: Sum scores

The manuscript treats sum scores as a measurement case. Data are generated on a latent continuous scale with no x-by-group interaction, converted into ordinal item responses, and summed. The atlas compares the latent generating scale, the observed sum-score identity model, and the Gaussian-probit bounded-score sensitivity model. It extends the manuscript scenarios across sample size, item-threshold location, number of items, latent dispersion, and the joint values of the predictor and group coefficients. In the main-effect surface, the latent-scale predictor coefficients are 0.425, 0.850, 1.275 and the group coefficients are -1.35, -0.90, -0.45.

### 5.1 Manuscript anchors

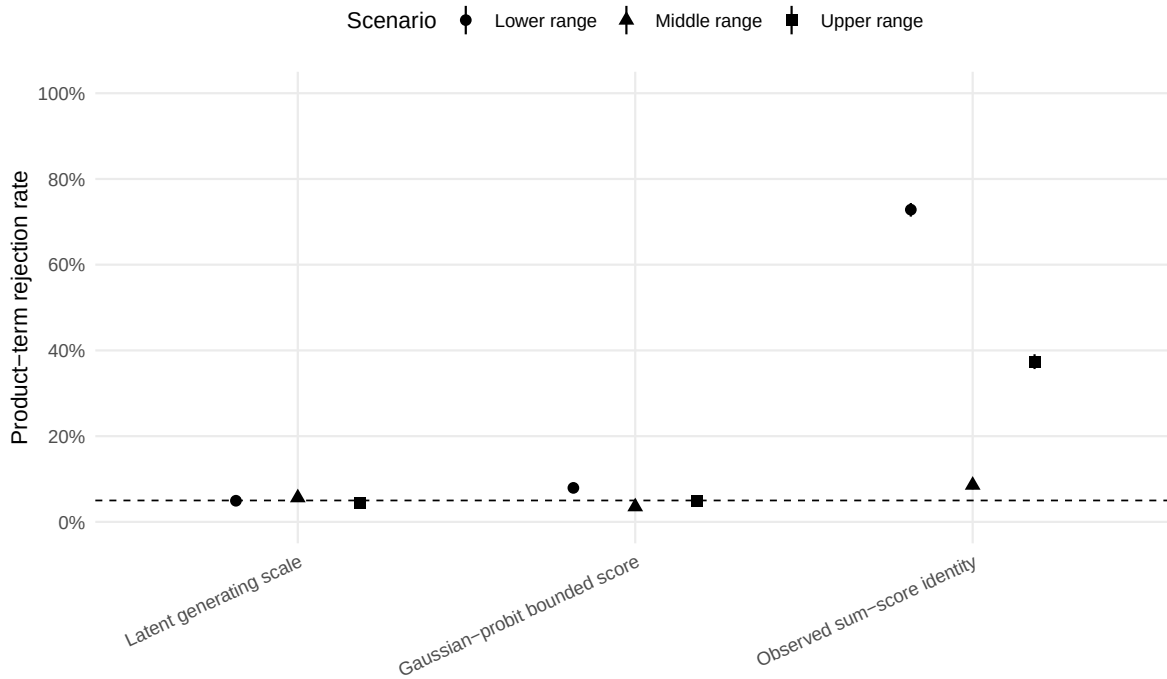


Figure 7: Sum-score manuscript anchors reproduced within the atlas. The product term is zero on the latent generating scale. The dashed line marks  $\alpha = .05$ .



Table 7: Sum-score manuscript anchors. The latent generating scale is available only in simulation.

| Scenario        | Generating<br>link/scale           | Fitted<br>model/scale                   | Rate type                                     | Rejection<br>rate | 95%<br>CI         | Deterministic<br>pseudo-<br>interaction | Fit<br>suc-<br>cess |
|-----------------|------------------------------------|-----------------------------------------|-----------------------------------------------|-------------------|-------------------|-----------------------------------------|---------------------|
| Lower<br>range  | graded<br>logistic<br>latent scale | Gaussian-<br>probit<br>bounded<br>score | Pseudo-<br>interaction<br>detection rate      | 7.9%              | [7.0%,<br>9.0%]   | -0.056                                  | 100.0%              |
| Lower<br>range  | graded<br>logistic<br>latent scale | Latent<br>generating<br>scale           | Nominal<br>generating-scale<br>rejection rate | 4.9%              | [4.2%,<br>5.8%]   | 0.000                                   | 100.0%              |
| Lower<br>range  | graded<br>logistic<br>latent scale | Observed<br>sum-score<br>identity       | Pseudo-<br>interaction<br>detection rate      | 72.8%             | [71.2%,<br>74.4%] | -2.034                                  | 100.0%              |
| Middle<br>range | graded<br>logistic<br>latent scale | Gaussian-<br>probit<br>bounded<br>score | Pseudo-<br>interaction<br>detection rate      | 3.5%              | [2.9%,<br>4.3%]   | -0.010                                  | 100.0%              |
| Middle<br>range | graded<br>logistic<br>latent scale | Latent<br>generating<br>scale           | Nominal<br>generating-scale<br>rejection rate | 5.6%              | [4.9%,<br>6.5%]   | 0.000                                   | 100.0%              |
| Middle<br>range | graded<br>logistic<br>latent scale | Observed<br>sum-score<br>identity       | Pseudo-<br>interaction<br>detection rate      | 8.6%              | [7.6%,<br>9.6%]   | -0.696                                  | 100.0%              |
| Upper<br>range  | graded<br>logistic<br>latent scale | Gaussian-<br>probit<br>bounded<br>score | Pseudo-<br>interaction<br>detection rate      | 4.8%              | [4.1%,<br>5.6%]   | 0.027                                   | 100.0%              |
| Upper<br>range  | graded<br>logistic<br>latent scale | Latent<br>generating<br>scale           | Nominal<br>generating-scale<br>rejection rate | 4.5%              | [3.8%,<br>5.3%]   | 0.000                                   | 100.0%              |
| Upper<br>range  | graded<br>logistic<br>latent scale | Observed<br>sum-score<br>identity       | Pseudo-<br>interaction<br>detection rate      | 37.4%             | [35.7%,<br>39.1%] | 1.570                                   | 100.0%              |

Across the atlas, the latent generating scale rejects from 4.2% to 6.0%. The observed sum-score identity model ranges from 2.4% to 82.1%, while the Gaussian-probit bounded-score sensitivity model ranges from 3.1% to 11.9%. The latter remains a model of the manifest total. It does

not recover the latent construct and should be read as a sensitivity analysis for the scale on which additivity is imposed.

## 5.2 One-dimensional sensitivity slices

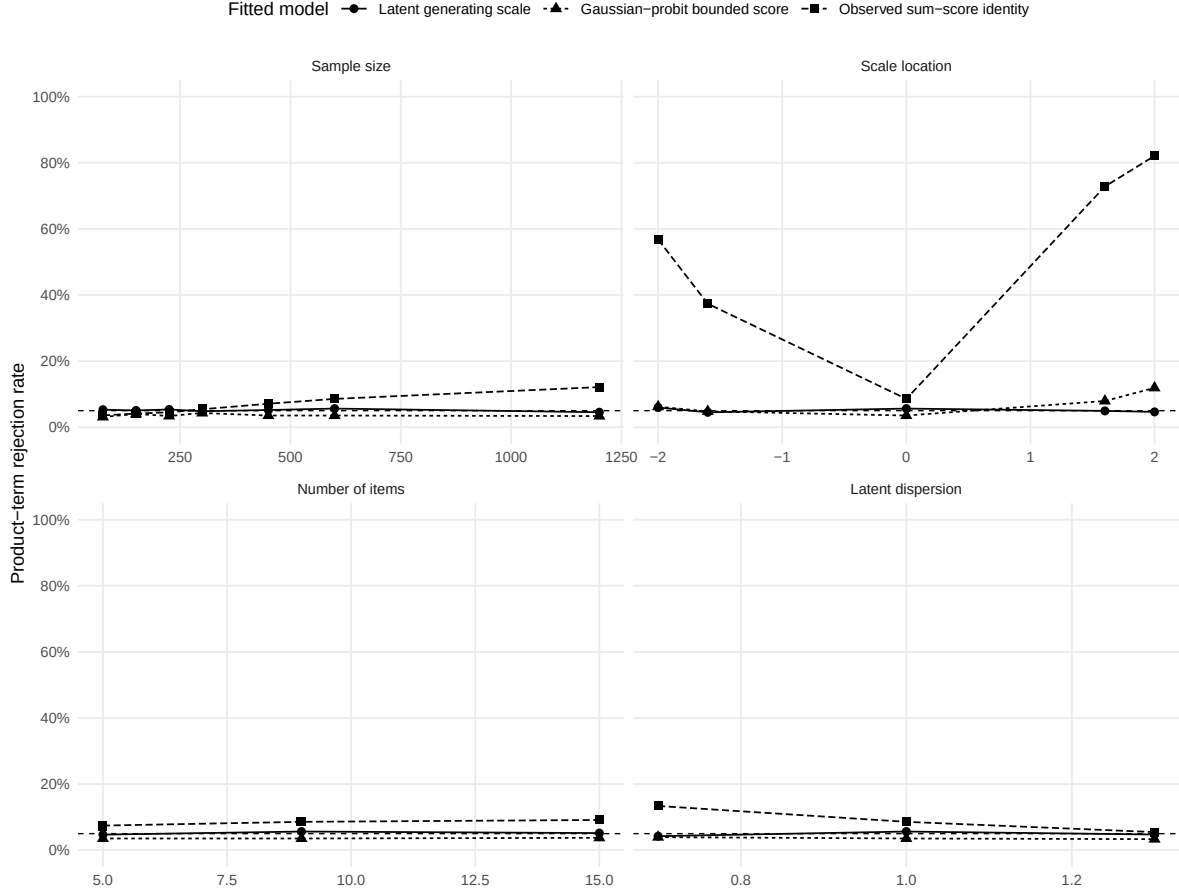


Figure 8: Sum-score sensitivity slices. One parameter varies at a time around the central manuscript anchor. The dashed line marks  $\alpha = .05$ .

Table 8: Range of rejection rates across each sum-score sensitivity slice.

| Slice             | Generating link/scale        | Fitted model/scale      | Minimum rejection rate | Maximum rejection rate |
|-------------------|------------------------------|-------------------------|------------------------|------------------------|
| Latent dispersion | graded logistic latent scale | Latent generating scale | 4.2%                   | 5.6%                   |

Table 8: Range of rejection rates across each sum-score sensitivity slice.

| Slice             | Generating link/scale        | Fitted model/scale            | Minimum rejection rate | Maximum rejection rate |
|-------------------|------------------------------|-------------------------------|------------------------|------------------------|
| Number of items   | graded logistic latent scale | Latent generating scale       | 4.7%                   | 5.6%                   |
| Sample size       | graded logistic latent scale | Latent generating scale       | 4.5%                   | 5.6%                   |
| Scale location    | graded logistic latent scale | Latent generating scale       | 4.5%                   | 6.0%                   |
| Latent dispersion | graded logistic latent scale | Gaussian-probit bounded score | 3.3%                   | 3.9%                   |
| Number of items   | graded logistic latent scale | Gaussian-probit bounded score | 3.5%                   | 3.7%                   |
| Sample size       | graded logistic latent scale | Gaussian-probit bounded score | 3.1%                   | 4.3%                   |
| Scale location    | graded logistic latent scale | Gaussian-probit bounded score | 3.5%                   | 11.9%                  |
| Latent dispersion | graded logistic latent scale | Observed sum-score identity   | 5.5%                   | 13.4%                  |
| Number of items   | graded logistic latent scale | Observed sum-score identity   | 7.4%                   | 9.1%                   |
| Sample size       | graded logistic latent scale | Observed sum-score identity   | 3.6%                   | 12.1%                  |
| Scale location    | graded logistic latent scale | Observed sum-score identity   | 8.6%                   | 82.1%                  |

The scale-location slice changes how strongly the bounded sum score compresses the latent dimension. Item count and latent dispersion change the mapping between the latent variable and the manifest total. Sample size changes the probability that any resulting fitted-scale interaction is detected. These are distinct components of the sum-score problem and should not be reduced to a generic floor-or-ceiling warning.

### 5.3 Main-effect surface

The surface shows how the mapping from latent values to a bounded total can turn two additive latent effects into a non-additive pattern on the manifest score. The latent benchmark remains the calibration reference because its product term is zero by construction.

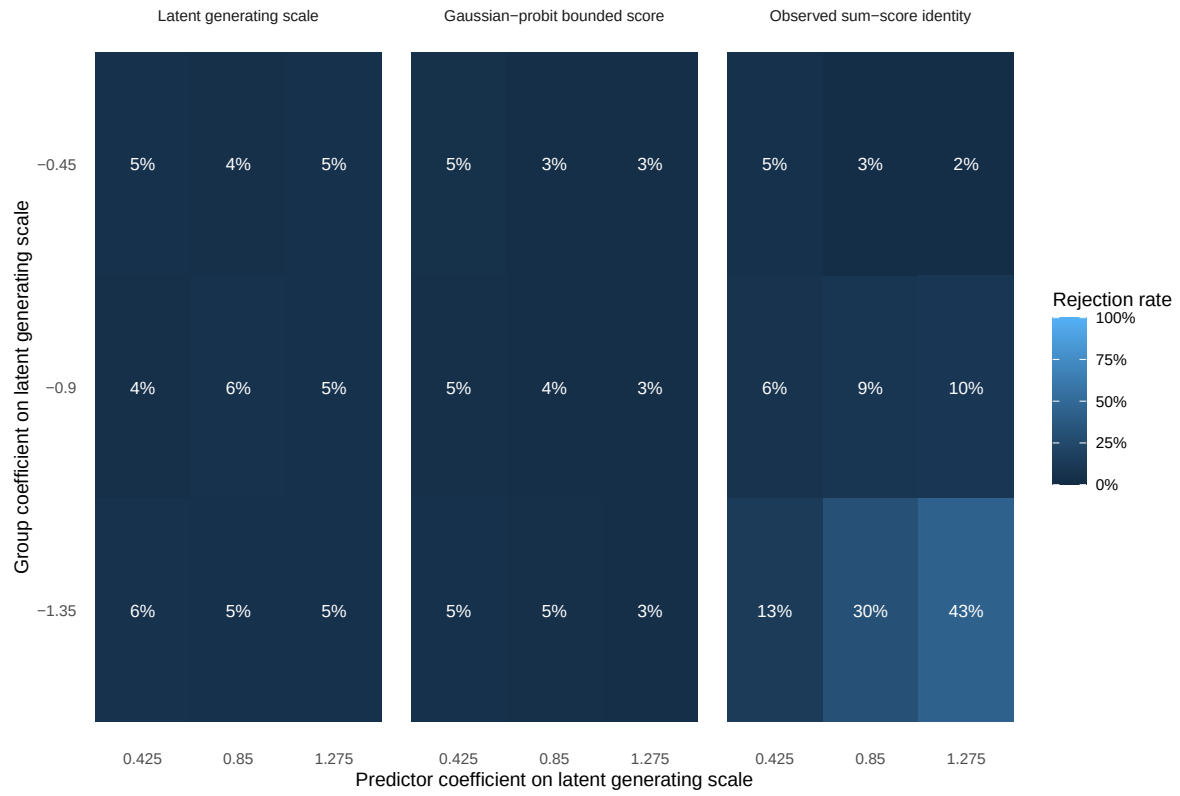


Figure 9: Sum-score main-effect surface. Axes report the predictor and group coefficients on the latent generating scale. Cell labels are product-term rejection rates.

## 6 Diagnostic atlas

The diagnostic atlas extends two of the manuscript’s five diagnostic scenarios: the forced-choice chance-floor case and the probit-generated, logit-fitted repeated-trial case. It is narrower than the main diagnostic simulation in the paper. Its purpose is to ask whether the relation between pseudo-interaction risk and diagnostic detection changes as sample size and the intercept vary.

Diagnostics are applied to the **misspecified interaction model**, matching the manuscript’s post-estimation question: once the fitted model has produced the product-term finding, does that same model show enough evidence of misspecification to warn the analyst? AIC compares prespecified candidate mean models with the same structural interaction formula. Positive delta AIC, defined as AIC of the misspecified model minus AIC of the generating model, favors the generating link.

Table 9: Diagnostic coverage in the current atlas summary. Applicable and computed are separate: a check can be structurally applicable but deferred in the current run.

| Case                                | Diagnostic                                       | Applicable | Computed |
|-------------------------------------|--------------------------------------------------|------------|----------|
| Forced-choice chance-floor case     | AIC favors generating link                       | Yes        | Yes      |
| Probit-generated, logit-fitted case | AIC favors generating link                       | Yes        | Yes      |
| Forced-choice chance-floor case     | DHARMa dispersion                                | Yes        | Yes      |
| Probit-generated, logit-fitted case | DHARMa dispersion                                | Yes        | Yes      |
| Forced-choice chance-floor case     | DHARMa residual distribution across design cells | No         | No       |
| Probit-generated, logit-fitted case | DHARMa residual distribution across design cells | Yes        | Yes      |
| Forced-choice chance-floor case     | DHARMa residual quantiles over fitted values     | Yes        | Yes      |
| Probit-generated, logit-fitted case | DHARMa residual quantiles over fitted values     | No         | No       |
| Forced-choice chance-floor case     | DHARMa residual quantiles over focal predictor   | Yes        | Yes      |
| Probit-generated, logit-fitted case | DHARMa residual quantiles over focal predictor   | No         | No       |

Table 9: Diagnostic coverage in the current atlas summary. Applicable and computed are separate: a check can be structurally applicable but deferred in the current run.

| Case                                | Diagnostic                           | Applicable | Computed |
|-------------------------------------|--------------------------------------|------------|----------|
| Forced-choice chance-floor case     | DHARMa uniformity                    | Yes        | Yes      |
| Probit-generated, logit-fitted case | DHARMa uniformity                    | Yes        | Yes      |
| Forced-choice chance-floor case     | Pregibon-style added-term link check | Yes        | Yes      |
| Probit-generated, logit-fitted case | Pregibon-style added-term link check | Yes        | Yes      |

The current diagnostic summary includes the applicable DHARMa checks.

## 6.1 Detection across sensitivity slices

The rates in this figure answer different questions. Pseudo-interaction detection is the probability that the misspecified fitted scale yields a significant product term. Pregibon-style detection is the probability that the added-term link check flags a departure. AIC reports how often the generating link has the lower AIC within the specified candidate pair. The curves therefore need not be complements of one another. Diagnostic misalignment occurs when the interaction conclusion changes more readily than the diagnostic workflow signals a problem.

## 6.2 Strength of the AIC comparison

Table 10: AIC comparison across the diagnostic atlas. Positive delta AIC favors the generating link. The within-two column reports the proportion of finite comparisons with absolute delta AIC below 2.

| Case                            | Setting           | Pseudo-interaction detection | AIC favors generating link | Median delta AIC | Delta AIC IQR | Absolute delta AIC < 2 |
|---------------------------------|-------------------|------------------------------|----------------------------|------------------|---------------|------------------------|
| Forced-choice chance-floor case | Sample size = 100 | 27.1%                        | 65.4%                      | 0.77             | [-0.53, 2.15] | 64.3%                  |

Table 10: AIC comparison across the diagnostic atlas. Positive delta AIC favors the generating link. The within-two column reports the proportion of finite comparisons with absolute delta AIC below 2.

| Case                                | Setting            | Pseudo-interaction detection | AIC favors generating link | Median delta AIC | Delta AIC IQR  | Absolute delta AIC < 2 |
|-------------------------------------|--------------------|------------------------------|----------------------------|------------------|----------------|------------------------|
| Forced-choice chance-floor case     | Sample size = 250  | 59.3%                        | 77.1%                      | 2.25             | [0.22, 4.55]   | 38.4%                  |
| Forced-choice chance-floor case     | Sample size = 500  | 86.9%                        | 85.4%                      | 4.44             | [1.67, 7.66]   | 21.0%                  |
| Forced-choice chance-floor case     | Sample size = 1000 | 99.1%                        | 94.7%                      | 9.62             | [5.56, 13.83]  | 6.7%                   |
| Probit-generated, logit-fitted case | Sample size = 200  | 7.8%                         | 91.6%                      | 3.92             | [1.99, 5.77]   | 23.1%                  |
| Probit-generated, logit-fitted case | Sample size = 300  | 10.3%                        | 95.4%                      | 6.04             | [3.72, 8.46]   | 11.1%                  |
| Probit-generated, logit-fitted case | Sample size = 700  | 18.5%                        | 99.3%                      | 13.76            | [10.02, 17.48] | 1.3%                   |
| Probit-generated, logit-fitted case | Sample size = 1200 | 26.9%                        | 100.0%                     | 23.54            | [18.83, 28.28] | 0.1%                   |
| Forced-choice chance-floor case     | Intercept = -0.80  | 59.3%                        | 77.1%                      | 2.25             | [0.22, 4.55]   | 38.4%                  |
| Forced-choice chance-floor case     | Intercept = -1.50  | 42.1%                        | 71.6%                      | 1.51             | [-0.24, 3.43]  | 47.9%                  |
| Forced-choice chance-floor case     | Intercept = 0.00   | 54.9%                        | 78.9%                      | 2.24             | [0.39, 4.35]   | 39.6%                  |
| Forced-choice chance-floor case     | Intercept = 0.80   | 33.6%                        | 72.5%                      | 1.36             | [-0.16, 2.94]  | 53.9%                  |
| Forced-choice chance-floor case     | Intercept = 1.50   | 13.7%                        | 65.4%                      | 0.55             | [-0.40, 1.58]  | 77.8%                  |

Table 10: AIC comparison across the diagnostic atlas. Positive delta AIC favors the generating link. The within-two column reports the proportion of finite comparisons with absolute delta AIC below 2.

| Case                                | Setting          | Pseudo-interaction detection | AIC favors generating link | Median delta AIC | Delta AIC IQR | Absolute delta AIC < 2 |
|-------------------------------------|------------------|------------------------------|----------------------------|------------------|---------------|------------------------|
| Probit-generated, logit-fitted case | Intercept = 1.50 | 10.3%                        | 95.4%                      | 6.04             | [3.72, 8.46]  | 11.1%                  |
| Probit-generated, logit-fitted case | Intercept = 0.00 | 19.8%                        | 95.3%                      | 6.68             | [4.03, 9.34]  | 10.3%                  |
| Probit-generated, logit-fitted case | Intercept = 0.75 | 6.1%                         | 95.9%                      | 5.56             | [3.34, 7.70]  | 12.9%                  |
| Probit-generated, logit-fitted case | Intercept = 2.25 | 22.1%                        | 93.9%                      | 6.50             | [3.58, 9.41]  | 12.3%                  |

The AIC win rate is reported together with the magnitude of the difference because these describe different situations. A generating link that wins frequently by a very small margin is weaker evidence than one that wins by a large margin. As in the manuscript, these comparisons rank a limited candidate set. They do not identify the substantively correct scale by themselves.

## 7 What the atlas adds, and what it does not

The atlas strengthens the simulation evidence in three ways. First, it shows whether the manuscript anchors sit inside broader patterns or depend narrowly on one tuned configuration. Second, it separates parameters that change the **deterministic scale discrepancy**, such as the intercept and main-effect magnitude, from parameters that mainly change **detectability**, such as sample size and trial count. Third, the diagnostic extension places pseudo-interaction detection and model-checking performance on the same sensitivity slices.

The atlas does not estimate how often link misspecification occurs in psychological research. It does not cover every plausible outcome family, link, random-effects structure, measurement model, or predictor distribution. It also does not turn model diagnostics into a decision rule for the scientifically correct scale. Its role is narrower: to show how the mechanisms developed in the paper change across a declared set of nearby scenarios.



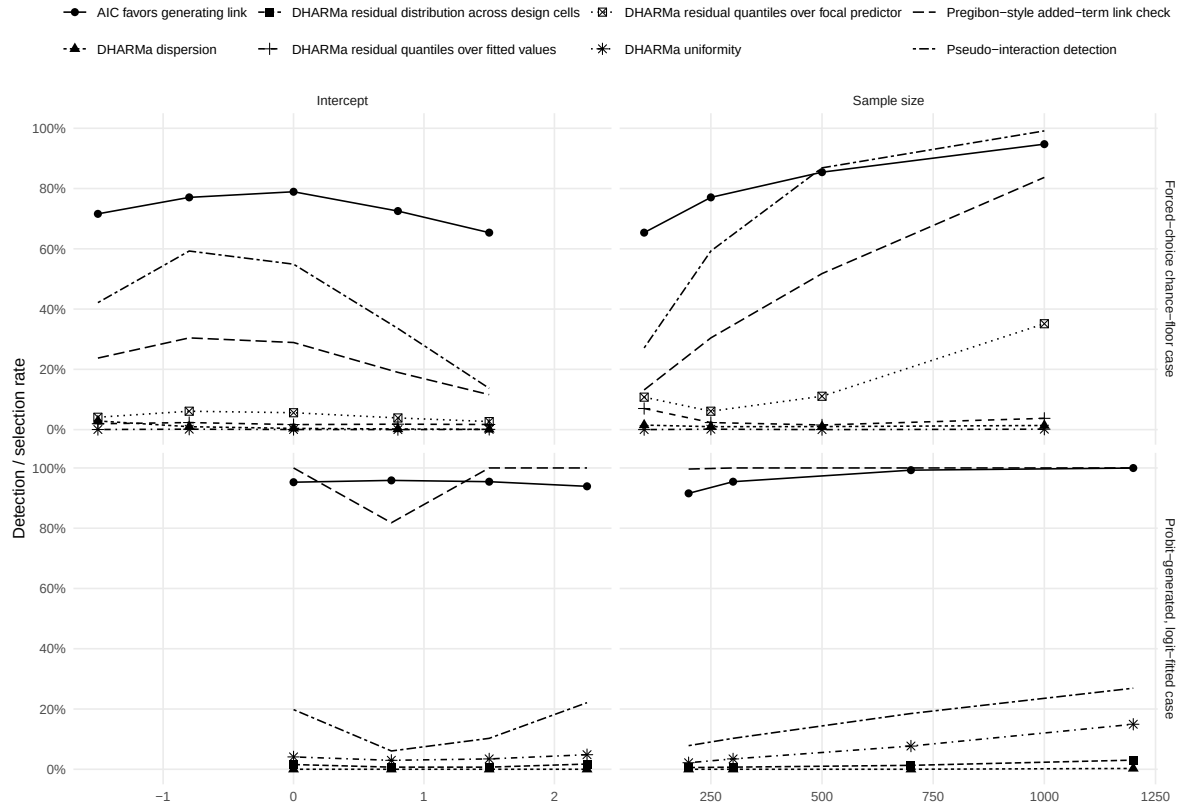


Figure 10: Diagnostic atlas. Pseudo-interaction detection is shown together with each diagnostic or candidate-model comparison computed in the current run. Sample-size and scale-location slices are shown separately.

The Shiny companion can be used for optional exploration of these results. This document is the stable reporting layer: every figure and table is generated from the compact atlas summaries stored in `simulation-atlas/data/`.

## 8 Numerical appendix

The tables below provide a compact scenario-level record of the core atlas. The machine-readable CSV files remain the authoritative full output and contain additional design, Monte Carlo, convergence, and response-scale columns.

### 8.1 Forced-choice accuracy

Table 11: Compact scenario-level results for the forced-choice atlas.

| ID     | Slice        | Scenario           | Generating link/scale  | Fitted model/scale              | Rejection rate | Deterministic pseudo-interaction | Fit success |
|--------|--------------|--------------------|------------------------|---------------------------------|----------------|----------------------------------|-------------|
| FC-001 | paper anchor | Lower performance  | chance-corrected logit | Chance-corrected binomial logit | 4.2%           | 0.000                            | 100.0%      |
| FC-001 | paper anchor | Lower performance  | chance-corrected logit | Gaussian identity               | 45.1%          | -0.077                           | 100.0%      |
| FC-001 | paper anchor | Lower performance  | chance-corrected logit | Standard binomial logit         | 59.3%          | -0.453                           | 100.0%      |
| FC-001 | paper anchor | Lower performance  | chance-corrected logit | Standard binomial probit        | 57.2%          | -0.262                           | 100.0%      |
| FC-002 | paper anchor | Middle performance | chance-corrected logit | Chance-corrected binomial logit | 5.8%           | 0.000                            | 100.0%      |
| FC-002 | paper anchor | Middle performance | chance-corrected logit | Gaussian identity               | 14.9%          | -0.036                           | 100.0%      |

Table 11: Compact scenario-level results for the forced-choice atlas.

| ID     | Slice               | Scenario                               | Generating link/scale  | Fitted model/scale              | Rejection rate | Deterministic pseudo-interaction | Fit success |
|--------|---------------------|----------------------------------------|------------------------|---------------------------------|----------------|----------------------------------|-------------|
| FC-002 | paper anchor        | Middle performance                     | chance-corrected logit | Standard binomial logit         | 55.7%          | -0.473                           | 100.0%      |
| FC-002 | paper anchor        | Middle performance                     | chance-corrected logit | Standard binomial probit        | 47.9%          | -0.243                           | 100.0%      |
| FC-003 | paper anchor        | Higher performance                     | chance-corrected logit | Chance-corrected binomial logit | 4.6%           | 0.000                            | 100.0%      |
| FC-003 | paper anchor        | Higher performance                     | chance-corrected logit | Gaussian identity               | 12.7%          | 0.028                            | 100.0%      |
| FC-003 | paper anchor        | Higher performance                     | chance-corrected logit | Standard binomial logit         | 33.2%          | -0.396                           | 100.0%      |
| FC-003 | paper anchor        | Higher performance                     | chance-corrected logit | Standard binomial probit        | 18.6%          | -0.153                           | 100.0%      |
| FC-004 | main effect surface | Age effect x 0.50; group effect x 0.50 | chance-corrected logit | Chance-corrected binomial logit | 4.3%           | 0.000                            | 100.0%      |
| FC-004 | main effect surface | Age effect x 0.50; group effect x 0.50 | chance-corrected logit | Gaussian identity               | 4.8%           | -0.007                           | 100.0%      |
| FC-004 | main effect surface | Age effect x 0.50; group effect x 0.50 | chance-corrected logit | Standard binomial logit         | 8.0%           | -0.124                           | 100.0%      |
| FC-004 | main effect surface | Age effect x 0.50; group effect x 0.50 | chance-corrected logit | Standard binomial probit        | 7.0%           | -0.063                           | 100.0%      |
| FC-005 | main effect surface | Age effect x 1.00; group effect x 0.50 | chance-corrected logit | Chance-corrected binomial logit | 5.0%           | 0.000                            | 100.0%      |

Table 11: Compact scenario-level results for the forced-choice atlas.

| ID     | Slice               | Scenario                               | Generating link/scale  | Fitted model/scale              | Rejection rate | Deterministic pseudo-interaction | Fit success |
|--------|---------------------|----------------------------------------|------------------------|---------------------------------|----------------|----------------------------------|-------------|
| FC-005 | main effect surface | Age effect x 1.00; group effect x 0.50 | chance-corrected logit | Gaussian identity               | 4.7%           | -0.009                           | 100.0%      |
| FC-005 | main effect surface | Age effect x 1.00; group effect x 0.50 | chance-corrected logit | Standard binomial logit         | 16.7%          | -0.232                           | 100.0%      |
| FC-005 | main effect surface | Age effect x 1.00; group effect x 0.50 | chance-corrected logit | Standard binomial probit        | 13.7%          | -0.113                           | 100.0%      |
| FC-006 | main effect surface | Age effect x 1.50; group effect x 0.50 | chance-corrected logit | Chance-corrected binomial logit | 5.1%           | 0.000                            | 100.0%      |
| FC-006 | main effect surface | Age effect x 1.50; group effect x 0.50 | chance-corrected logit | Gaussian identity               | 5.3%           | -0.009                           | 100.0%      |
| FC-006 | main effect surface | Age effect x 1.50; group effect x 0.50 | chance-corrected logit | Standard binomial logit         | 26.1%          | -0.313                           | 100.0%      |
| FC-006 | main effect surface | Age effect x 1.50; group effect x 0.50 | chance-corrected logit | Standard binomial probit        | 20.4%          | -0.145                           | 100.0%      |
| FC-007 | main effect surface | Age effect x 0.50; group effect x 1.00 | chance-corrected logit | Chance-corrected binomial logit | 4.2%           | 0.000                            | 100.0%      |
| FC-007 | main effect surface | Age effect x 0.50; group effect x 1.00 | chance-corrected logit | Gaussian identity               | 8.1%           | -0.024                           | 100.0%      |
| FC-007 | main effect surface | Age effect x 0.50; group effect x 1.00 | chance-corrected logit | Standard binomial logit         | 20.8%          | -0.255                           | 100.0%      |
| FC-007 | main effect surface | Age effect x 0.50; group effect x 1.00 | chance-corrected logit | Standard binomial probit        | 17.6%          | -0.136                           | 100.0%      |

Table 11: Compact scenario-level results for the forced-choice atlas.

| ID     | Slice               | Scenario                               | Generating link/scale  | Fitted model/scale              | Rejection rate | Deterministic pseudo-interaction | Fit success |
|--------|---------------------|----------------------------------------|------------------------|---------------------------------|----------------|----------------------------------|-------------|
| FC-008 | main effect surface | Age effect x 1.50; group effect x 1.00 | chance-corrected logit | Chance-corrected binomial logit | 5.1%           | 0.000                            | 100.0%      |
| FC-008 | main effect surface | Age effect x 1.50; group effect x 1.00 | chance-corrected logit | Gaussian identity               | 20.1%          | -0.034                           | 100.0%      |
| FC-008 | main effect surface | Age effect x 1.50; group effect x 1.00 | chance-corrected logit | Standard binomial logit         | 78.3%          | -0.635                           | 100.0%      |
| FC-008 | main effect surface | Age effect x 1.50; group effect x 1.00 | chance-corrected logit | Standard binomial probit        | 69.6%          | -0.309                           | 100.0%      |
| FC-009 | main effect surface | Age effect x 0.50; group effect x 1.50 | chance-corrected logit | Chance-corrected binomial logit | 4.6%           | 0.000                            | 100.0%      |
| FC-009 | main effect surface | Age effect x 0.50; group effect x 1.50 | chance-corrected logit | Gaussian identity               | 19.4%          | -0.048                           | 100.0%      |
| FC-009 | main effect surface | Age effect x 0.50; group effect x 1.50 | chance-corrected logit | Standard binomial logit         | 41.1%          | -0.380                           | 100.0%      |
| FC-009 | main effect surface | Age effect x 0.50; group effect x 1.50 | chance-corrected logit | Standard binomial probit        | 36.1%          | -0.210                           | 100.0%      |
| FC-010 | main effect surface | Age effect x 1.00; group effect x 1.50 | chance-corrected logit | Chance-corrected binomial logit | 4.6%           | 0.000                            | 100.0%      |
| FC-010 | main effect surface | Age effect x 1.00; group effect x 1.50 | chance-corrected logit | Gaussian identity               | 47.0%          | -0.073                           | 100.0%      |
| FC-010 | main effect surface | Age effect x 1.00; group effect x 1.50 | chance-corrected logit | Standard binomial logit         | 88.8%          | -0.706                           | 100.0%      |

Table 11: Compact scenario-level results for the forced-choice atlas.

| ID     | Slice               | Scenario                               | Generating link/scale  | Fitted model/scale              | Rejection rate | Deterministic pseudo-interaction | Fit success |
|--------|---------------------|----------------------------------------|------------------------|---------------------------------|----------------|----------------------------------|-------------|
| FC-010 | main effect surface | Age effect x 1.00; group effect x 1.50 | chance-corrected logit | Standard binomial probit        | 84.0%          | -0.376                           | 100.0%      |
| FC-011 | main effect surface | Age effect x 1.50; group effect x 1.50 | chance-corrected logit | Chance-corrected binomial logit | 3.7%           | 0.000                            | 100.0%      |
| FC-011 | main effect surface | Age effect x 1.50; group effect x 1.50 | chance-corrected logit | Gaussian identity               | 60.7%          | -0.073                           | 100.0%      |
| FC-011 | main effect surface | Age effect x 1.50; group effect x 1.50 | chance-corrected logit | Standard binomial logit         | 98.6%          | -0.949                           | 100.0%      |
| FC-011 | main effect surface | Age effect x 1.50; group effect x 1.50 | chance-corrected logit | Standard binomial probit        | 97.1%          | -0.481                           | 100.0%      |
| FC-012 | sample size         | N = 50                                 | chance-corrected logit | Chance-corrected binomial logit | 4.3%           | 0.000                            | 100.0%      |
| FC-012 | sample size         | N = 50                                 | chance-corrected logit | Gaussian identity               | 6.5%           | -0.036                           | 100.0%      |
| FC-012 | sample size         | N = 50                                 | chance-corrected logit | Standard binomial logit         | 15.2%          | -0.473                           | 100.0%      |
| FC-012 | sample size         | N = 50                                 | chance-corrected logit | Standard binomial probit        | 13.8%          | -0.243                           | 100.0%      |
| FC-013 | sample size         | N = 100                                | chance-corrected logit | Chance-corrected binomial logit | 4.7%           | 0.000                            | 100.0%      |
| FC-013 | sample size         | N = 100                                | chance-corrected logit | Gaussian identity               | 9.3%           | -0.036                           | 100.0%      |

Table 11: Compact scenario-level results for the forced-choice atlas.

| ID     | Slice       | Scenario | Generating<br>link/scale | Fitted<br>model/scale           | Rejection<br>rate | Deterministic<br>pseudo-<br>interaction | Fit<br>suc-<br>cess |
|--------|-------------|----------|--------------------------|---------------------------------|-------------------|-----------------------------------------|---------------------|
| FC-013 | sample size | N = 100  | chance-corrected logit   | Standard binomial logit         | 27.5%             | -0.473                                  | 100.0%              |
| FC-013 | sample size | N = 100  | chance-corrected logit   | Standard binomial probit        | 23.3%             | -0.243                                  | 100.0%              |
| FC-014 | sample size | N = 150  | chance-corrected logit   | Chance-corrected binomial logit | 4.9%              | 0.000                                   | 100.0%              |
| FC-014 | sample size | N = 150  | chance-corrected logit   | Gaussian identity               | 11.1%             | -0.036                                  | 100.0%              |
| FC-014 | sample size | N = 150  | chance-corrected logit   | Standard binomial logit         | 36.5%             | -0.473                                  | 100.0%              |
| FC-014 | sample size | N = 150  | chance-corrected logit   | Standard binomial probit        | 30.8%             | -0.243                                  | 100.0%              |
| FC-015 | sample size | N = 200  | chance-corrected logit   | Chance-corrected binomial logit | 4.7%              | 0.000                                   | 100.0%              |
| FC-015 | sample size | N = 200  | chance-corrected logit   | Gaussian identity               | 13.0%             | -0.036                                  | 100.0%              |
| FC-015 | sample size | N = 200  | chance-corrected logit   | Standard binomial logit         | 47.3%             | -0.473                                  | 100.0%              |
| FC-015 | sample size | N = 200  | chance-corrected logit   | Standard binomial probit        | 39.4%             | -0.243                                  | 100.0%              |
| FC-016 | sample size | N = 500  | chance-corrected logit   | Chance-corrected binomial logit | 5.2%              | 0.000                                   | 100.0%              |

Table 11: Compact scenario-level results for the forced-choice atlas.

| ID     | Slice          | Scenario          | Generating<br>link/scale | Fitted<br>model/scale           | Rejection<br>rate | Deterministic<br>pseudo-<br>interaction | Fit<br>suc-<br>cess |
|--------|----------------|-------------------|--------------------------|---------------------------------|-------------------|-----------------------------------------|---------------------|
| FC-016 | sample size    | N = 500           | chance-corrected logit   | Gaussian identity               | 27.3%             | -0.036                                  | 100.0%              |
| FC-016 | sample size    | N = 500           | chance-corrected logit   | Standard binomial logit         | 85.7%             | -0.473                                  | 100.0%              |
| FC-016 | sample size    | N = 500           | chance-corrected logit   | Standard binomial probit        | 78.5%             | -0.243                                  | 100.0%              |
| FC-017 | sample size    | N = 1000          | chance-corrected logit   | Chance-corrected binomial logit | 4.8%              | 0.000                                   | 100.0%              |
| FC-017 | sample size    | N = 1000          | chance-corrected logit   | Gaussian identity               | 47.2%             | -0.036                                  | 100.0%              |
| FC-017 | sample size    | N = 1000          | chance-corrected logit   | Standard binomial logit         | 99.0%             | -0.473                                  | 100.0%              |
| FC-017 | sample size    | N = 1000          | chance-corrected logit   | Standard binomial probit        | 97.2%             | -0.243                                  | 100.0%              |
| FC-018 | scale location | Intercept = -1.50 | chance-corrected logit   | Chance-corrected binomial logit | 4.6%              | 0.000                                   | 100.0%              |
| FC-018 | scale location | Intercept = -1.50 | chance-corrected logit   | Gaussian identity               | 39.2%             | -0.079                                  | 100.0%              |
| FC-018 | scale location | Intercept = -1.50 | chance-corrected logit   | Standard binomial logit         | 43.8%             | -0.365                                  | 100.0%              |
| FC-018 | scale location | Intercept = -1.50 | chance-corrected logit   | Standard binomial probit        | 43.2%             | -0.222                                  | 100.0%              |



Table 11: Compact scenario-level results for the forced-choice atlas.

| ID     | Slice             | Scenario         | Generating<br>link/scale      | Fitted<br>model/scale                     | Rejection<br>rate | Deterministic<br>pseudo-<br>interaction | Fit<br>suc-<br>cess |
|--------|-------------------|------------------|-------------------------------|-------------------------------------------|-------------------|-----------------------------------------|---------------------|
| FC-019 | scale<br>location | Intercept = 1.50 | chance-<br>corrected<br>logit | Chance-<br>corrected<br>binomial<br>logit | 5.0%              | 0.000                                   | 100.0%              |
| FC-019 | scale<br>location | Intercept = 1.50 | chance-<br>corrected<br>logit | Gaussian<br>identity                      | 65.5%             | 0.071                                   | 100.0%              |
| FC-019 | scale<br>location | Intercept = 1.50 | chance-<br>corrected<br>logit | Standard<br>binomial<br>logit             | 14.0%             | -0.286                                  | 100.0%              |
| FC-019 | scale<br>location | Intercept = 1.50 | chance-<br>corrected<br>logit | Standard<br>binomial<br>probit            | 6.3%              | -0.053                                  | 100.0%              |
| FC-020 | trial<br>count    | Trials = 5       | chance-<br>corrected<br>logit | Chance-<br>corrected<br>binomial<br>logit | 3.6%              | 0.000                                   | 100.0%              |
| FC-020 | trial<br>count    | Trials = 5       | chance-<br>corrected<br>logit | Gaussian<br>identity                      | 7.1%              | -0.036                                  | 100.0%              |
| FC-020 | trial<br>count    | Trials = 5       | chance-<br>corrected<br>logit | Standard<br>binomial<br>logit             | 18.0%             | -0.473                                  | 100.0%              |
| FC-020 | trial<br>count    | Trials = 5       | chance-<br>corrected<br>logit | Standard<br>binomial<br>probit            | 15.7%             | -0.243                                  | 100.0%              |
| FC-021 | trial<br>count    | Trials = 10      | chance-<br>corrected<br>logit | Chance-<br>corrected<br>binomial<br>logit | 4.9%              | 0.000                                   | 100.0%              |
| FC-021 | trial<br>count    | Trials = 10      | chance-<br>corrected<br>logit | Gaussian<br>identity                      | 10.6%             | -0.036                                  | 100.0%              |
| FC-021 | trial<br>count    | Trials = 10      | chance-<br>corrected<br>logit | Standard<br>binomial<br>logit             | 32.3%             | -0.473                                  | 100.0%              |

Table 11: Compact scenario-level results for the forced-choice atlas.

| ID     | Slice           | Scenario           | Generating<br>link/scale      | Fitted<br>model/scale                     | Rejection<br>rate | Deterministic<br>pseudo-<br>interaction | Fit<br>suc-<br>cess |
|--------|-----------------|--------------------|-------------------------------|-------------------------------------------|-------------------|-----------------------------------------|---------------------|
| FC-021 | trial<br>count  | Trials = 10        | chance-<br>corrected<br>logit | Standard<br>binomial<br>probit            | 27.3%             | -0.243                                  | 100.0%              |
| FC-022 | trial<br>count  | Trials = 50        | chance-<br>corrected<br>logit | Chance-<br>corrected<br>binomial<br>logit | 5.0%              | 0.000                                   | 100.0%              |
| FC-022 | trial<br>count  | Trials = 50        | chance-<br>corrected<br>logit | Gaussian<br>identity                      | 30.5%             | -0.036                                  | 100.0%              |
| FC-022 | trial<br>count  | Trials = 50        | chance-<br>corrected<br>logit | Standard<br>binomial<br>logit             | 90.5%             | -0.473                                  | 100.0%              |
| FC-022 | trial<br>count  | Trials = 50        | chance-<br>corrected<br>logit | Standard<br>binomial<br>probit            | 83.8%             | -0.243                                  | 100.0%              |
| FC-023 | chance<br>level | Chance = 0.25      | chance-<br>corrected<br>logit | Chance-<br>corrected<br>binomial<br>logit | 5.2%              | 0.000                                   | 100.0%              |
| FC-023 | chance<br>level | Chance = 0.25      | chance-<br>corrected<br>logit | Gaussian<br>identity                      | 25.7%             | -0.054                                  | 100.0%              |
| FC-023 | chance<br>level | Chance = 0.25      | chance-<br>corrected<br>logit | Standard<br>binomial<br>logit             | 47.5%             | -0.411                                  | 100.0%              |
| FC-023 | chance<br>level | Chance = 0.25      | chance-<br>corrected<br>logit | Standard<br>binomial<br>probit            | 44.7%             | -0.228                                  | 100.0%              |
| FC-024 | chance<br>level | Chance =<br>0.3333 | chance-<br>corrected<br>logit | Chance-<br>corrected<br>binomial<br>logit | 4.6%              | 0.000                                   | 100.0%              |
| FC-024 | chance<br>level | Chance =<br>0.3333 | chance-<br>corrected<br>logit | Gaussian<br>identity                      | 21.2%             | -0.048                                  | 100.0%              |

Table 11: Compact scenario-level results for the forced-choice atlas.

| ID     | Slice        | Scenario        | Generating link/scale  | Fitted model/scale       | Rejection rate | Deterministic pseudo-interaction | Fit success |
|--------|--------------|-----------------|------------------------|--------------------------|----------------|----------------------------------|-------------|
| FC-024 | chance level | Chance = 0.3333 | chance-corrected logit | Standard binomial logit  | 54.3%          | -0.444                           | 100.0%      |
| FC-024 | chance level | Chance = 0.3333 | chance-corrected logit | Standard binomial probit | 48.2%          | -0.240                           | 100.0%      |

## 8.2 Logit versus probit

Table 12: Compact scenario-level results for the within-family logit-probit atlas.

| ID     | Slice               | Scenario                                        | Generating link/scale | Fitted model/scale | Rejection rate | Deterministic pseudo-interaction | Fit success |
|--------|---------------------|-------------------------------------------------|-----------------------|--------------------|----------------|----------------------------------|-------------|
| WF-001 | paper anchor        | Logit-generated manuscript scenario             | logit                 | Binomial logit     | 5.6%           | 0.000                            | 100.0%      |
| WF-001 | paper anchor        | Logit-generated manuscript scenario             | logit                 | Binomial probit    | 16.5%          | -0.112                           | 100.0%      |
| WF-002 | paper anchor        | Probit-generated manuscript scenario            | probit                | Binomial logit     | 17.6%          | 0.216                            | 100.0%      |
| WF-002 | paper anchor        | Probit-generated manuscript scenario            | probit                | Binomial probit    | 5.0%           | 0.000                            | 100.0%      |
| WF-003 | main effect surface | logit generated: group x 0.50; condition x 0.50 | logit                 | Binomial logit     | 4.5%           | 0.000                            | 100.0%      |
| WF-003 | main effect surface | logit generated: group x 0.50; condition x 0.50 | logit                 | Binomial probit    | 7.4%           | -0.048                           | 100.0%      |
| WF-004 | main effect surface | logit generated: group x 1.00; condition x 0.50 | logit                 | Binomial logit     | 4.9%           | 0.000                            | 100.0%      |
| WF-004 | main effect surface | logit generated: group x 1.00; condition x 0.50 | logit                 | Binomial probit    | 10.3%          | -0.080                           | 100.0%      |

Table 12: Compact scenario-level results for the within-family logit-probit atlas.

| ID     | Slice                     | Scenario                                              | Generating<br>link/scale | Fitted<br>model/scale | Rejection<br>rate | Deterministic<br>pseudo-<br>interaction | Fit<br>suc-<br>cess |
|--------|---------------------------|-------------------------------------------------------|--------------------------|-----------------------|-------------------|-----------------------------------------|---------------------|
| WF-005 | main<br>effect<br>surface | logit generated:<br>group x 1.50;<br>condition x 0.50 | logit                    | Binomial<br>logit     | 4.7%              | 0.000                                   | 100.0%              |
| WF-005 | main<br>effect<br>surface | logit generated:<br>group x 1.50;<br>condition x 0.50 | logit                    | Binomial<br>probit    | 11.7%             | -0.080                                  | 100.0%              |
| WF-006 | main<br>effect<br>surface | logit generated:<br>group x 0.50;<br>condition x 1.00 | logit                    | Binomial<br>logit     | 5.3%              | 0.000                                   | 100.0%              |
| WF-006 | main<br>effect<br>surface | logit generated:<br>group x 0.50;<br>condition x 1.00 | logit                    | Binomial<br>probit    | 11.8%             | -0.080                                  | 100.0%              |
| WF-007 | main<br>effect<br>surface | logit generated:<br>group x 1.50;<br>condition x 1.00 | logit                    | Binomial<br>logit     | 5.6%              | 0.000                                   | 100.0%              |
| WF-007 | main<br>effect<br>surface | logit generated:<br>group x 1.50;<br>condition x 1.00 | logit                    | Binomial<br>probit    | 11.2%             | -0.080                                  | 100.0%              |
| WF-008 | main<br>effect<br>surface | logit generated:<br>group x 0.50;<br>condition x 1.50 | logit                    | Binomial<br>logit     | 5.2%              | 0.000                                   | 100.0%              |
| WF-008 | main<br>effect<br>surface | logit generated:<br>group x 0.50;<br>condition x 1.50 | logit                    | Binomial<br>probit    | 10.2%             | -0.080                                  | 100.0%              |
| WF-009 | main<br>effect<br>surface | logit generated:<br>group x 1.00;<br>condition x 1.50 | logit                    | Binomial<br>logit     | 5.2%              | 0.000                                   | 100.0%              |
| WF-009 | main<br>effect<br>surface | logit generated:<br>group x 1.00;<br>condition x 1.50 | logit                    | Binomial<br>probit    | 10.6%             | -0.080                                  | 100.0%              |
| WF-010 | main<br>effect<br>surface | logit generated:<br>group x 1.50;<br>condition x 1.50 | logit                    | Binomial<br>logit     | 5.1%              | 0.000                                   | 100.0%              |
| WF-010 | main<br>effect<br>surface | logit generated:<br>group x 1.50;<br>condition x 1.50 | logit                    | Binomial<br>probit    | 5.8%              | 0.000                                   | 100.0%              |

Table 12: Compact scenario-level results for the within-family logit-probit atlas.

| ID     | Slice                     | Scenario                                               | Generating<br>link/scale | Fitted<br>model/scale | Rejection<br>rate | Deterministic<br>pseudo-<br>interaction | Fit<br>suc-<br>cess |
|--------|---------------------------|--------------------------------------------------------|--------------------------|-----------------------|-------------------|-----------------------------------------|---------------------|
| WF-011 | main<br>effect<br>surface | probit generated:<br>group x 0.50;<br>condition x 0.50 | probit                   | Binomial<br>logit     | 8.5%              | 0.107                                   | 100.0%              |
| WF-011 | main<br>effect<br>surface | probit generated:<br>group x 0.50;<br>condition x 0.50 | probit                   | Binomial<br>probit    | 5.7%              | 0.000                                   | 100.0%              |
| WF-012 | main<br>effect<br>surface | probit generated:<br>group x 1.00;<br>condition x 0.50 | probit                   | Binomial<br>logit     | 12.4%             | 0.162                                   | 100.0%              |
| WF-012 | main<br>effect<br>surface | probit generated:<br>group x 1.00;<br>condition x 0.50 | probit                   | Binomial<br>probit    | 5.2%              | 0.000                                   | 100.0%              |
| WF-013 | main<br>effect<br>surface | probit generated:<br>group x 1.50;<br>condition x 0.50 | probit                   | Binomial<br>logit     | 13.2%             | 0.162                                   | 100.0%              |
| WF-013 | main<br>effect<br>surface | probit generated:<br>group x 1.50;<br>condition x 0.50 | probit                   | Binomial<br>probit    | 5.3%              | 0.000                                   | 100.0%              |
| WF-014 | main<br>effect<br>surface | probit generated:<br>group x 0.50;<br>condition x 1.00 | probit                   | Binomial<br>logit     | 11.2%             | 0.162                                   | 100.0%              |
| WF-014 | main<br>effect<br>surface | probit generated:<br>group x 0.50;<br>condition x 1.00 | probit                   | Binomial<br>probit    | 4.9%              | 0.000                                   | 100.0%              |
| WF-015 | main<br>effect<br>surface | probit generated:<br>group x 1.50;<br>condition x 1.00 | probit                   | Binomial<br>logit     | 12.1%             | 0.162                                   | 100.0%              |
| WF-015 | main<br>effect<br>surface | probit generated:<br>group x 1.50;<br>condition x 1.00 | probit                   | Binomial<br>probit    | 5.1%              | 0.000                                   | 100.0%              |
| WF-016 | main<br>effect<br>surface | probit generated:<br>group x 0.50;<br>condition x 1.50 | probit                   | Binomial<br>logit     | 10.9%             | 0.162                                   | 100.0%              |
| WF-016 | main<br>effect<br>surface | probit generated:<br>group x 0.50;<br>condition x 1.50 | probit                   | Binomial<br>probit    | 4.8%              | 0.000                                   | 100.0%              |

Table 12: Compact scenario-level results for the within-family logit-probit atlas.

| ID     | Slice               | Scenario                                               | Generating<br>link/scale | Fitted<br>model/scale | Rejection<br>rate | Deterministic<br>pseudo-<br>interaction | Fit<br>suc-<br>cess |
|--------|---------------------|--------------------------------------------------------|--------------------------|-----------------------|-------------------|-----------------------------------------|---------------------|
| WF-017 | main effect surface | probit generated:<br>group x 1.00;<br>condition x 1.50 | probit                   | Binomial<br>logit     | 10.2%             | 0.162                                   | 100.0%              |
| WF-017 | main effect surface | probit generated:<br>group x 1.00;<br>condition x 1.50 | probit                   | Binomial<br>probit    | 5.2%              | 0.000                                   | 100.0%              |
| WF-018 | main effect surface | probit generated:<br>group x 1.50;<br>condition x 1.50 | probit                   | Binomial<br>logit     | 4.4%              | 0.000                                   | 100.0%              |
| WF-018 | main effect surface | probit generated:<br>group x 1.50;<br>condition x 1.50 | probit                   | Binomial<br>probit    | 4.7%              | 0.000                                   | 100.0%              |
| WF-019 | sample size         | logit generated:<br>subjects = 100                     | logit                    | Binomial<br>logit     | 5.5%              | 0.000                                   | 100.0%              |
| WF-019 | sample size         | logit generated:<br>subjects = 100                     | logit                    | Binomial<br>probit    | 7.4%              | -0.112                                  | 100.0%              |
| WF-020 | sample size         | logit generated:<br>subjects = 200                     | logit                    | Binomial<br>logit     | 5.6%              | 0.000                                   | 100.0%              |
| WF-020 | sample size         | logit generated:<br>subjects = 200                     | logit                    | Binomial<br>probit    | 9.3%              | -0.112                                  | 100.0%              |
| WF-021 | sample size         | logit generated:<br>subjects = 300                     | logit                    | Binomial<br>logit     | 5.7%              | 0.000                                   | 100.0%              |
| WF-021 | sample size         | logit generated:<br>subjects = 300                     | logit                    | Binomial<br>probit    | 11.9%             | -0.112                                  | 100.0%              |
| WF-022 | sample size         | logit generated:<br>subjects = 400                     | logit                    | Binomial<br>logit     | 4.4%              | 0.000                                   | 100.0%              |
| WF-022 | sample size         | logit generated:<br>subjects = 400                     | logit                    | Binomial<br>probit    | 11.8%             | -0.112                                  | 100.0%              |
| WF-023 | sample size         | logit generated:<br>subjects = 1200                    | logit                    | Binomial<br>logit     | 5.3%              | 0.000                                   | 100.0%              |
| WF-023 | sample size         | logit generated:<br>subjects = 1200                    | logit                    | Binomial<br>probit    | 25.9%             | -0.112                                  | 100.0%              |
| WF-024 | sample size         | logit generated:<br>subjects = 2000                    | logit                    | Binomial<br>logit     | 4.7%              | 0.000                                   | 100.0%              |

Table 12: Compact scenario-level results for the within-family logit-probit atlas.

| ID     | Slice           | Scenario                                       | Generating<br>link/scale | Fitted<br>model/scale | Rejection<br>rate | Deterministic<br>pseudo-<br>interaction | Fit<br>suc-<br>cess |
|--------|-----------------|------------------------------------------------|--------------------------|-----------------------|-------------------|-----------------------------------------|---------------------|
| WF-024 | sample size     | logit generated:<br>subjects = 2000            | logit                    | Binomial<br>probit    | 38.5%             | -0.112                                  | 100.0%              |
| WF-025 | trial count     | logit generated:<br>trials per cell = 5        | logit                    | Binomial<br>logit     | 4.8%              | 0.000                                   | 100.0%              |
| WF-025 | trial count     | logit generated:<br>trials per cell = 5        | logit                    | Binomial<br>probit    | 8.2%              | -0.112                                  | 100.0%              |
| WF-026 | trial count     | logit generated:<br>trials per cell = 30       | logit                    | Binomial<br>logit     | 4.4%              | 0.000                                   | 100.0%              |
| WF-026 | trial count     | logit generated:<br>trials per cell = 30       | logit                    | Binomial<br>probit    | 31.0%             | -0.112                                  | 100.0%              |
| WF-027 | latent location | logit generated:<br>reference intercept = 0    | logit                    | Binomial<br>logit     | 5.4%              | 0.000                                   | 100.0%              |
| WF-027 | latent location | logit generated:<br>reference intercept = 0    | logit                    | Binomial<br>probit    | 38.2%             | 0.175                                   | 100.0%              |
| WF-028 | latent location | logit generated:<br>reference intercept = 0.75 | logit                    | Binomial<br>logit     | 5.1%              | 0.000                                   | 100.0%              |
| WF-028 | latent location | logit generated:<br>reference intercept = 0.75 | logit                    | Binomial<br>probit    | 8.2%              | 0.059                                   | 100.0%              |
| WF-029 | latent location | logit generated:<br>reference intercept = 2.25 | logit                    | Binomial<br>logit     | 5.6%              | 0.000                                   | 100.0%              |
| WF-029 | latent location | logit generated:<br>reference intercept = 2.25 | logit                    | Binomial<br>probit    | 44.1%             | -0.186                                  | 100.0%              |
| WF-030 | icc             | logit generated:<br>target ICC = 0             | logit                    | Binomial<br>logit     | 4.9%              | 0.000                                   | 100.0%              |
| WF-030 | icc             | logit generated:<br>target ICC = 0             | logit                    | Binomial<br>probit    | 78.7%             | -0.112                                  | 100.0%              |
| WF-031 | icc             | logit generated:<br>target ICC = 0.15          | logit                    | Binomial<br>logit     | 5.5%              | 0.000                                   | 100.0%              |

Table 12: Compact scenario-level results for the within-family logit-probit atlas.

| ID     | Slice          | Scenario                                 | Generating<br>link/scale | Fitted<br>model/scale | Rejection<br>rate | Deterministic<br>pseudo-<br>interaction | Fit<br>suc-<br>cess |
|--------|----------------|------------------------------------------|--------------------------|-----------------------|-------------------|-----------------------------------------|---------------------|
| WF-031 | icc            | logit generated:<br>target ICC = 0.15    | logit                    | Binomial<br>probit    | 37.0%             | -0.112                                  | 100.0%              |
| WF-032 | icc            | logit generated:<br>target ICC = 0.45    | logit                    | Binomial<br>logit     | 5.9%              | 0.000                                   | 100.0%              |
| WF-032 | icc            | logit generated:<br>target ICC = 0.45    | logit                    | Binomial<br>probit    | 9.2%              | -0.112                                  | 100.0%              |
| WF-033 | sample<br>size | probit generated:<br>subjects = 100      | probit                   | Binomial<br>logit     | 6.6%              | 0.216                                   | 100.0%              |
| WF-033 | sample<br>size | probit generated:<br>subjects = 100      | probit                   | Binomial<br>probit    | 5.3%              | 0.000                                   | 100.0%              |
| WF-034 | sample<br>size | probit generated:<br>subjects = 200      | probit                   | Binomial<br>logit     | 8.8%              | 0.216                                   | 100.0%              |
| WF-034 | sample<br>size | probit generated:<br>subjects = 200      | probit                   | Binomial<br>probit    | 5.5%              | 0.000                                   | 100.0%              |
| WF-035 | sample<br>size | probit generated:<br>subjects = 300      | probit                   | Binomial<br>logit     | 11.5%             | 0.216                                   | 100.0%              |
| WF-035 | sample<br>size | probit generated:<br>subjects = 300      | probit                   | Binomial<br>probit    | 5.6%              | 0.000                                   | 100.0%              |
| WF-036 | sample<br>size | probit generated:<br>subjects = 400      | probit                   | Binomial<br>logit     | 11.6%             | 0.216                                   | 100.0%              |
| WF-036 | sample<br>size | probit generated:<br>subjects = 400      | probit                   | Binomial<br>probit    | 4.5%              | 0.000                                   | 100.0%              |
| WF-037 | sample<br>size | probit generated:<br>subjects = 1200     | probit                   | Binomial<br>logit     | 28.1%             | 0.216                                   | 100.0%              |
| WF-037 | sample<br>size | probit generated:<br>subjects = 1200     | probit                   | Binomial<br>probit    | 5.5%              | 0.000                                   | 100.0%              |
| WF-038 | sample<br>size | probit generated:<br>subjects = 2000     | probit                   | Binomial<br>logit     | 41.5%             | 0.216                                   | 100.0%              |
| WF-038 | sample<br>size | probit generated:<br>subjects = 2000     | probit                   | Binomial<br>probit    | 5.6%              | 0.000                                   | 100.0%              |
| WF-039 | trial<br>count | probit generated:<br>trials per cell = 5 | probit                   | Binomial<br>logit     | 7.9%              | 0.216                                   | 100.0%              |
| WF-039 | trial<br>count | probit generated:<br>trials per cell = 5 | probit                   | Binomial<br>probit    | 4.8%              | 0.000                                   | 100.0%              |



Table 12: Compact scenario-level results for the within-family logit-probit atlas.

| ID     | Slice              | Scenario                                        | Generating<br>link/scale | Fitted<br>model/scale | Rejection<br>rate | Deterministic<br>pseudo-<br>interaction | Fit<br>suc-<br>cess |
|--------|--------------------|-------------------------------------------------|--------------------------|-----------------------|-------------------|-----------------------------------------|---------------------|
| WF-040 | trial<br>count     | probit generated:<br>trials per cell = 30       | probit                   | Binomial<br>logit     | 32.2%             | 0.216                                   | 100.0%              |
| WF-040 | trial<br>count     | probit generated:<br>trials per cell = 30       | probit                   | Binomial<br>probit    | 4.6%              | 0.000                                   | 100.0%              |
| WF-041 | latent<br>location | probit generated:<br>reference intercept = 0    | probit                   | Binomial<br>logit     | 39.9%             | -0.424                                  | 100.0%              |
| WF-041 | latent<br>location | probit generated:<br>reference intercept = 0    | probit                   | Binomial<br>probit    | 5.7%              | 0.000                                   | 100.0%              |
| WF-042 | latent<br>location | probit generated:<br>reference intercept = 0.75 | probit                   | Binomial<br>logit     | 8.0%              | -0.108                                  | 100.0%              |
| WF-042 | latent<br>location | probit generated:<br>reference intercept = 0.75 | probit                   | Binomial<br>probit    | 4.7%              | 0.000                                   | 100.0%              |
| WF-043 | latent<br>location | probit generated:<br>reference intercept = 2.25 | probit                   | Binomial<br>logit     | 46.2%             | 0.520                                   | 100.0%              |
| WF-043 | latent<br>location | probit generated:<br>reference intercept = 2.25 | probit                   | Binomial<br>probit    | 4.7%              | 0.000                                   | 100.0%              |
| WF-044 | icc                | probit generated:<br>target ICC = 0             | probit                   | Binomial<br>logit     | 82.3%             | 0.216                                   | 100.0%              |
| WF-044 | icc                | probit generated:<br>target ICC = 0             | probit                   | Binomial<br>probit    | 5.0%              | 0.000                                   | 100.0%              |
| WF-045 | icc                | probit generated:<br>target ICC = 0.15          | probit                   | Binomial<br>logit     | 40.9%             | 0.216                                   | 100.0%              |
| WF-045 | icc                | probit generated:<br>target ICC = 0.15          | probit                   | Binomial<br>probit    | 4.8%              | 0.000                                   | 100.0%              |
| WF-046 | icc                | probit generated:<br>target ICC = 0.45          | probit                   | Binomial<br>logit     | 8.6%              | 0.216                                   | 100.0%              |
| WF-046 | icc                | probit generated:<br>target ICC = 0.45          | probit                   | Binomial<br>probit    | 5.4%              | 0.000                                   | 100.0%              |

### 8.3 Sum scores

Table 13: Compact scenario-level results for the sum-score atlas.

| ID     | Slice               | Scenario                             | Generating link/scale        | Fitted model/scale            | Rejection rate | Deterministic pseudo-interaction | Fit success |
|--------|---------------------|--------------------------------------|------------------------------|-------------------------------|----------------|----------------------------------|-------------|
| SS-001 | paper anchor        | Lower range                          | graded logistic latent scale | Gaussian-probit bounded score | 7.9%           | -0.056                           | 100.0%      |
| SS-001 | paper anchor        | Lower range                          | graded logistic latent scale | Latent generating scale       | 4.9%           | 0.000                            | 100.0%      |
| SS-001 | paper anchor        | Lower range                          | graded logistic latent scale | Observed sum-score identity   | 72.8%          | -2.034                           | 100.0%      |
| SS-002 | paper anchor        | Middle range                         | graded logistic latent scale | Gaussian-probit bounded score | 3.5%           | -0.010                           | 100.0%      |
| SS-002 | paper anchor        | Middle range                         | graded logistic latent scale | Latent generating scale       | 5.6%           | 0.000                            | 100.0%      |
| SS-002 | paper anchor        | Middle range                         | graded logistic latent scale | Observed sum-score identity   | 8.6%           | -0.696                           | 100.0%      |
| SS-003 | paper anchor        | Upper range                          | graded logistic latent scale | Gaussian-probit bounded score | 4.8%           | 0.027                            | 100.0%      |
| SS-003 | paper anchor        | Upper range                          | graded logistic latent scale | Latent generating scale       | 4.5%           | 0.000                            | 100.0%      |
| SS-003 | paper anchor        | Upper range                          | graded logistic latent scale | Observed sum-score identity   | 37.4%          | 1.570                            | 100.0%      |
| SS-004 | main effect surface | x effect x 0.50; group effect x 0.50 | graded logistic latent scale | Gaussian-probit bounded score | 4.7%           | -0.001                           | 100.0%      |

Table 13: Compact scenario-level results for the sum-score atlas.

| ID     | Slice               | Scenario                                | Generating link/scale        | Fitted model/scale            | Rejection rate | Deterministic pseudo-interaction | Fit success |
|--------|---------------------|-----------------------------------------|------------------------------|-------------------------------|----------------|----------------------------------|-------------|
| SS-004 | main effect surface | x effect x 0.50;<br>group effect x 0.50 | graded logistic latent scale | Latent generating scale       | 4.9%           | 0.000                            | 100.0%      |
| SS-004 | main effect surface | x effect x 0.50;<br>group effect x 0.50 | graded logistic latent scale | Observed sum-score identity   | 4.8%           | -0.096                           | 100.0%      |
| SS-005 | main effect surface | x effect x 1.00;<br>group effect x 0.50 | graded logistic latent scale | Gaussian-probit bounded score | 3.3%           | -0.002                           | 100.0%      |
| SS-005 | main effect surface | x effect x 1.00;<br>group effect x 0.50 | graded logistic latent scale | Latent generating scale       | 4.3%           | 0.000                            | 100.0%      |
| SS-005 | main effect surface | x effect x 1.00;<br>group effect x 0.50 | graded logistic latent scale | Observed sum-score identity   | 2.6%           | -0.180                           | 100.0%      |
| SS-006 | main effect surface | x effect x 1.50;<br>group effect x 0.50 | graded logistic latent scale | Gaussian-probit bounded score | 3.2%           | -0.004                           | 100.0%      |
| SS-006 | main effect surface | x effect x 1.50;<br>group effect x 0.50 | graded logistic latent scale | Latent generating scale       | 4.8%           | 0.000                            | 100.0%      |
| SS-006 | main effect surface | x effect x 1.50;<br>group effect x 0.50 | graded logistic latent scale | Observed sum-score identity   | 2.4%           | -0.242                           | 100.0%      |
| SS-007 | main effect surface | x effect x 0.50;<br>group effect x 1.00 | graded logistic latent scale | Gaussian-probit bounded score | 4.6%           | -0.005                           | 100.0%      |
| SS-007 | main effect surface | x effect x 0.50;<br>group effect x 1.00 | graded logistic latent scale | Latent generating scale       | 4.5%           | 0.000                            | 100.0%      |
| SS-007 | main effect surface | x effect x 0.50;<br>group effect x 1.00 | graded logistic latent scale | Observed sum-score identity   | 6.4%           | -0.370                           | 100.0%      |

Table 13: Compact scenario-level results for the sum-score atlas.

| ID     | Slice               | Scenario                             | Generating link/scale        | Fitted model/scale            | Rejection rate | Deterministic pseudo-interaction | Fit success |
|--------|---------------------|--------------------------------------|------------------------------|-------------------------------|----------------|----------------------------------|-------------|
| SS-008 | main effect surface | x effect x 1.50; group effect x 1.00 | graded logistic latent scale | Gaussian-probit bounded score | 3.5%           | -0.015                           | 100.0%      |
| SS-008 | main effect surface | x effect x 1.50; group effect x 1.00 | graded logistic latent scale | Latent generating scale       | 5.4%           | 0.000                            | 100.0%      |
| SS-008 | main effect surface | x effect x 1.50; group effect x 1.00 | graded logistic latent scale | Observed sum-score identity   | 10.5%          | -0.940                           | 100.0%      |
| SS-009 | main effect surface | x effect x 0.50; group effect x 1.50 | graded logistic latent scale | Gaussian-probit bounded score | 4.9%           | -0.011                           | 100.0%      |
| SS-009 | main effect surface | x effect x 0.50; group effect x 1.50 | graded logistic latent scale | Latent generating scale       | 5.6%           | 0.000                            | 100.0%      |
| SS-009 | main effect surface | x effect x 0.50; group effect x 1.50 | graded logistic latent scale | Observed sum-score identity   | 13.3%          | -0.786                           | 100.0%      |
| SS-010 | main effect surface | x effect x 1.00; group effect x 1.50 | graded logistic latent scale | Gaussian-probit bounded score | 4.5%           | -0.022                           | 100.0%      |
| SS-010 | main effect surface | x effect x 1.00; group effect x 1.50 | graded logistic latent scale | Latent generating scale       | 5.1%           | 0.000                            | 100.0%      |
| SS-010 | main effect surface | x effect x 1.00; group effect x 1.50 | graded logistic latent scale | Observed sum-score identity   | 30.0%          | -1.481                           | 100.0%      |
| SS-011 | main effect surface | x effect x 1.50; group effect x 1.50 | graded logistic latent scale | Gaussian-probit bounded score | 3.5%           | -0.036                           | 100.0%      |

Table 13: Compact scenario-level results for the sum-score atlas.

| ID     | Slice               | Scenario                             | Generating link/scale        | Fitted model/scale            | Rejection rate | Deterministic pseudo-interaction | Fit success |
|--------|---------------------|--------------------------------------|------------------------------|-------------------------------|----------------|----------------------------------|-------------|
| SS-011 | main effect surface | x effect x 1.50; group effect x 1.50 | graded logistic latent scale | Latent generating scale       | 4.9%           | 0.000                            | 100.0%      |
| SS-011 | main effect surface | x effect x 1.50; group effect x 1.50 | graded logistic latent scale | Observed sum-score identity   | 42.9%          | -2.008                           | 100.0%      |
| SS-012 | sample size         | N = 75                               | graded logistic latent scale | Gaussian-probit bounded score | 3.1%           | -0.010                           | 100.0%      |
| SS-012 | sample size         | N = 75                               | graded logistic latent scale | Latent generating scale       | 5.3%           | 0.000                            | 100.0%      |
| SS-012 | sample size         | N = 75                               | graded logistic latent scale | Observed sum-score identity   | 3.6%           | -0.696                           | 100.0%      |
| SS-013 | sample size         | N = 150                              | graded logistic latent scale | Gaussian-probit bounded score | 4.0%           | -0.010                           | 100.0%      |
| SS-013 | sample size         | N = 150                              | graded logistic latent scale | Latent generating scale       | 5.1%           | 0.000                            | 100.0%      |
| SS-013 | sample size         | N = 150                              | graded logistic latent scale | Observed sum-score identity   | 4.1%           | -0.696                           | 100.0%      |
| SS-014 | sample size         | N = 225                              | graded logistic latent scale | Gaussian-probit bounded score | 3.4%           | -0.010                           | 100.0%      |
| SS-014 | sample size         | N = 225                              | graded logistic latent scale | Latent generating scale       | 5.4%           | 0.000                            | 100.0%      |
| SS-014 | sample size         | N = 225                              | graded logistic latent scale | Observed sum-score identity   | 4.5%           | -0.696                           | 100.0%      |

Table 13: Compact scenario-level results for the sum-score atlas.

| ID     | Slice          | Scenario                | Generating<br>link/scale     | Fitted<br>model/scale         | Rejection<br>rate | Deterministic<br>pseudo-<br>interaction | Fit<br>suc-<br>cess |
|--------|----------------|-------------------------|------------------------------|-------------------------------|-------------------|-----------------------------------------|---------------------|
| SS-015 | sample size    | N = 300                 | graded logistic latent scale | Gaussian-probit bounded score | 4.3%              | -0.010                                  | 100.0%              |
| SS-015 | sample size    | N = 300                 | graded logistic latent scale | Latent generating scale       | 4.8%              | 0.000                                   | 100.0%              |
| SS-015 | sample size    | N = 300                 | graded logistic latent scale | Observed sum-score identity   | 5.4%              | -0.696                                  | 100.0%              |
| SS-016 | sample size    | N = 450                 | graded logistic latent scale | Gaussian-probit bounded score | 3.5%              | -0.010                                  | 100.0%              |
| SS-016 | sample size    | N = 450                 | graded logistic latent scale | Latent generating scale       | 5.2%              | 0.000                                   | 100.0%              |
| SS-016 | sample size    | N = 450                 | graded logistic latent scale | Observed sum-score identity   | 7.1%              | -0.696                                  | 100.0%              |
| SS-017 | sample size    | N = 1200                | graded logistic latent scale | Gaussian-probit bounded score | 3.4%              | -0.010                                  | 100.0%              |
| SS-017 | sample size    | N = 1200                | graded logistic latent scale | Latent generating scale       | 4.5%              | 0.000                                   | 100.0%              |
| SS-017 | sample size    | N = 1200                | graded logistic latent scale | Observed sum-score identity   | 12.1%             | -0.696                                  | 100.0%              |
| SS-018 | scale location | Threshold shift = -2.00 | graded logistic latent scale | Gaussian-probit bounded score | 6.2%              | 0.038                                   | 100.0%              |

Table 13: Compact scenario-level results for the sum-score atlas.

| ID     | Slice             | Scenario                   | Generating<br>link/scale           | Fitted<br>model/scale                   | Rejection<br>rate | Deterministic<br>pseudo-<br>interaction | Fit<br>suc-<br>cess |
|--------|-------------------|----------------------------|------------------------------------|-----------------------------------------|-------------------|-----------------------------------------|---------------------|
| SS-018 | scale<br>location | Threshold shift<br>= -2.00 | graded<br>logistic<br>latent scale | Latent<br>generating<br>scale           | 6.0%              | 0.000                                   | 100.0%              |
| SS-018 | scale<br>location | Threshold shift<br>= -2.00 | graded<br>logistic<br>latent scale | Observed<br>sum-score<br>identity       | 56.8%             | 1.877                                   | 100.0%              |
| SS-019 | scale<br>location | Threshold shift<br>= 2.00  | graded<br>logistic<br>latent scale | Gaussian-<br>probit<br>bounded<br>score | 11.9%             | -0.072                                  | 100.0%              |
| SS-019 | scale<br>location | Threshold shift<br>= 2.00  | graded<br>logistic<br>latent scale | Latent<br>generating<br>scale           | 4.6%              | 0.000                                   | 100.0%              |
| SS-019 | scale<br>location | Threshold shift<br>= 2.00  | graded<br>logistic<br>latent scale | Observed<br>sum-score<br>identity       | 82.1%             | -1.993                                  | 100.0%              |
| SS-020 | item<br>count     | Items = 5                  | graded<br>logistic<br>latent scale | Gaussian-<br>probit<br>bounded<br>score | 3.5%              | -0.013                                  | 100.0%              |
| SS-020 | item<br>count     | Items = 5                  | graded<br>logistic<br>latent scale | Latent<br>generating<br>scale           | 4.7%              | 0.000                                   | 100.0%              |
| SS-020 | item<br>count     | Items = 5                  | graded<br>logistic<br>latent scale | Observed<br>sum-score<br>identity       | 7.4%              | -0.384                                  | 100.0%              |
| SS-021 | item<br>count     | Items = 15                 | graded<br>logistic<br>latent scale | Gaussian-<br>probit<br>bounded<br>score | 3.7%              | -0.008                                  | 100.0%              |
| SS-021 | item<br>count     | Items = 15                 | graded<br>logistic<br>latent scale | Latent<br>generating<br>scale           | 5.2%              | 0.000                                   | 100.0%              |
| SS-021 | item<br>count     | Items = 15                 | graded<br>logistic<br>latent scale | Observed<br>sum-score<br>identity       | 9.1%              | -1.162                                  | 100.0%              |

Table 13: Compact scenario-level results for the sum-score atlas.

| ID     | Slice             | Scenario         | Generating<br>link/scale     | Fitted<br>model/scale         | Rejection<br>rate | Deterministic<br>pseudo-<br>interaction | Fit<br>suc-<br>cess |
|--------|-------------------|------------------|------------------------------|-------------------------------|-------------------|-----------------------------------------|---------------------|
| SS-022 | latent dispersion | Latent SD = 0.70 | graded logistic latent scale | Gaussian-probit bounded score | 3.9%              | -0.013                                  | 100.0%              |
| SS-022 | latent dispersion | Latent SD = 0.70 | graded logistic latent scale | Latent generating scale       | 4.2%              | 0.000                                   | 100.0%              |
| SS-022 | latent dispersion | Latent SD = 0.70 | graded logistic latent scale | Observed sum-score identity   | 13.4%             | -0.831                                  | 100.0%              |
| SS-023 | latent dispersion | Latent SD = 1.30 | graded logistic latent scale | Gaussian-probit bounded score | 3.3%              | -0.007                                  | 100.0%              |
| SS-023 | latent dispersion | Latent SD = 1.30 | graded logistic latent scale | Latent generating scale       | 4.7%              | 0.000                                   | 100.0%              |
| SS-023 | latent dispersion | Latent SD = 1.30 | graded logistic latent scale | Observed sum-score identity   | 5.5%              | -0.563                                  | 100.0%              |